Mapping Employee Satisfaction: Clustering, Prediction and Sentiment Insights

Build machine learning models to inform business decisions

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1 Business understanding

1.1 What are the business objectives?

Background

ServiceFirst is a national organisation that services large equipment for corporate and private clients in Australia. They are successful business with 25 years in the industry and their company boasts 250 employees within a multi-layered structure with various departments. This includes remote service staff and office-based support teams spread across regional offices nationwide.

Unfortunately, during the pandemic many employees resigned which led to high recruitment costs and a loss of experience within the organisation. Therefore, ServiceFirst initiated a project to better understand the reasons for the resignations.

The initial research project explored employee perceptions of the organisational climate using a questionnaire based on the Patterson Organisational Climate Measure (Patterson et al., 2005). Analysis using multiple linear regression revealed that factors such as tenure, decision-making involvement, autonomy, relationships with direct managers, team integration, welfare, training, work-life balance, and responsiveness to customers significantly influenced overall organisational climate (OC) scores. Among these, team integration had the strongest impact, followed by autonomy and positive relationships with direct managers. Additionally, employees with 13 years of tenure reported higher OC scores than those with 8 years, suggesting that longer tenure may foster stronger engagement and integration within the organisation.

A follow-up survey was conducted to further monitor changes in employee satisfaction and address any ongoing concerns. The following information was collected:

- Frequency of business travel,
- Level of education,
- Gender,
- Age,
- Monthly income,
- Total working years,
- Satisfaction score (based on the Organisational Climate Measure) (Patterson et al., 2005) and
- Open-ended feedback.

Business objectives and success criteria

ServiceFirst's overall aim is to reduce recruitment costs and minimise the loss of experienced employees by fostering a more positive organisational climate. Specifically, the goals of this second project is to gain a deeper understanding on the key drivers of employee satisfaction by:

1. Identifying clusters of employees with similar satisfaction levels and monthly income.
2. Creating a model to predict satisfaction levels based on various factors.
3. Conducting a sentiment analysis of open-ended feedback.

Additional consultation with ServiceFirst's Executive team is required to establish specific, measurable targets for employee retention and recruitment cost reduction, along with a defined project timeline.

1.2 Assess the situation

Inventory of resources

- Data scientist pro bono
- Python language within Jupyter notebook.
- Various libraries and modules were imported. In Jupyter they were only imported only once when they were first required (refer to Table 1).

Table 1 Libraries and modules imported

Libraries and modules imported	Description
pandas as pd NumPy as np	Numpy provides a variety of mathematical functions that can perform on entire array of data without the need for loops. Some of the applications of Numpy include data cleaning (e.g., fill missing data) and providing statistical analysis. An extension of Numpy, Pandas enables importing data from a file (e.g., csv) and in turn preview, manipulate, transform and visualise that data. Numpy and Pandas are typically imported together to perform initial cleaning and exploration of data.
warnings.filterwarnings('ignore')	Suppresses warning messages that alert for potential issues in the code that might not necessarily cause it to fail but could lead to problems under certain conditions.
matplotlib.pyplot as plt seaborn as sns	Utilised for creating visualisations. These libraries enable the creation of various charts and plots, including bar charts, box plots, histograms, and scatterplots, to help understand and explore data trends and distributions.
K-means clustering	
from sklearn.preprocessing import StandardScaler	Standardises the data to ensure features are on a consistent scale, which is critical for clustering algorithms
from sklearn.cluster import KMeans	Employed to perform the Elbow Method analysis, which helps determine the optimal number of clusters by analysing the within-cluster variance.
from sklearn.metrics import silhouette_score	Performs Silhouette analysis to evaluate cluster separation and further refine the choice of the optimal number of clusters.
from sklearn.metrics import v_measure_score	Use to calculate the v_measure which measures the balance of homogeneity and completeness of the clusters
from scipy.spatial.distance import cdist	Function to calculate cluster cardinality (i.e., the number of data points in each cluster)
Decision tree modelling	
from sklearn.model_selection import train_test_split	Used to split the dataset into training and testing subsets, enabling model validation.
from sklearn.tree import DecisionTreeClassifier	Perform decision tree modelling
from sklearn import tree from sklearn.tree import export_text	To creating a textual representation of the decision modelling tree
import graphviz from sklearn.tree import export_graphviz from IPython.display import display	To create a graphical representation of the decision modelling tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report	To evaluate the model's performance by providing metrics such as providing metrics such as accuracy, precision, and recall.
Natural language processing	
import contractions	Expands shortened forms (e.g., "can't" to "cannot") for better processing.

Libraries and modules imported	Description
import nltk nltk.download('punkt_tab') from nltk.tokenize import word_tokenize	Facilitates tokenisation, dividing text into individual words using the **nltk.tokenize.word_tokenize** function
from nltk.probability import FreqDist	To calculate the frequency distribution of the words
from nltk.stem import WordNetLemmatizer from nltk.corpus import wordnet from nltk import pos_tag nltk.download('averaged_perceptron_tagger_eng') nltk.download('wordnet')	To perform lemmatisation which is the process of reducing words to their base or root form (e.g., "running" becomes "run," and "better" becomes "good").
nltk.download ('stopwords') from nltk.corpus import stopwords	Used to remove stopwords (common words in a language e.g., "the," "and," "is")
from wordcloud import WordCloud	To perform wordclouds for visualising text data
from collections import Counter	To count the occurrences of elements in a collection (like a list, tuple, or string).
from textblob import TextBlob	To conduct sentiment analysis and determine the polarity (positivity/negativity) and subjectivity of a given text.
from nltk.sentiment.vader import SentimentIntensityAnalyzer nltk.download('vader_lexicon')	To conduct sentiment analysis using the Vader tool from the NLTK library

Assumptions

The P&C team's data collection process is assumed to have adhered to ethical research standards, ensuring the integrity of the study and the protection of participants. This likely included obtaining informed consent, maintaining confidentiality, and ensuring participation was entirely voluntary. Transparent communication about the purpose of the research would have been prioritised, allowing participants to make informed decisions about their involvement. Measures to securely manage sensitive information were likely in place to safeguard privacy, and participants would have been assured of their right to withdraw at any time without consequence. Adhering to these ethical guidelines is essential for maintaining trust and safeguarding the wellbeing of all participants.

1.3 Data mining goals

The data mining goals relevant to the business goals are provided in Table 2.

Table 2 Data mining goal against business goal

Business goal	Data mining goal
1: Identifying groups of employees with similar satisfaction levels and monthly incomes	Identify employee clusters by: • Using k-means clustering to group employees based on satisfaction levels and monthly income. • Understanding distinct clusters to tailor strategies for different employee segments.
2: Creating a model to predict satisfaction levels based on various factors	Predict satisfaction scores by: • Building and evaluating a decision tree model to predict overall satisfaction scores based on various factors. • Assessing the model's performance and identifying areas for improvement.
3: Conducting a sentiment analysis of open-ended feedback.	Conduct sentiment analysis by: • Preprocessing and tokenising employee feedback. • Creating visualisations and determine frequency distributions. • Reviewing the polarity and sentiment of the feedback to identify the top positive and negative sentiments.

1.4 Project plan

The data science project plan to achieve the business goals follows the Cross Industry Standard Process for Data Mining (CRISP-DM) framework and is provided in Table 3.

Table 3 Data science project plan

Plan	Description
1. Business understanding	• Define business objectives, assess the situation, and set goals for the data mining project (provided in Section 1)
2. Understand the data	• Collect and describe the data • Explore the data by performing descriptive statistics and visualisations to gain a better understanding of the key factors that impact on the overall OC measure • Examine the quality of the data • The initial exploration of the data and analyses will determine if there is any data transformation/preparation required (refer to Section 2)
3. Prepare and clean the data	• Select and clean the data to be used for analyses • Construct and combine data if required (refer to Section 3)

4. Conduct analysis, modelling and testing	• Conduct the following:
	  • k-means clustering
	  • decision tree modelling
	  • natural language processing (refer to Section 4)

5. Evaluate the analysis, provide recommendations and conclusion	• Evaluate the analysis to ensure they meet the business goals • Provide conclusions and recommendations based on initial data science project • Provide strategies for ServiceFirst to achieve the business goals (refer to Section 5)
6. Deployment	• Provide a written report (this report)

2 Data understanding

2.1 Initial data collection

ServiceFirst conducted a follow-up survey and a description of each of the items is provide in Table 4.

The structured data was provided in a comma-separated values (CSV) file named *employee_satisfaction_data.csv* and imported into a Jupyter Notebook for analysis in Python. The dataframe (df) was called 'survey'.

2.2 Data description

As shown below from the Jupyter notebook the original *employee_satisfaction_data.csv (survey df)* consists of 9 columns and 142 entries. The data set consists of three data types (object or numbers as either float64 and int64).

```
RangeIndex: 142 entries, 0 to 141
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Rowid               142 non-null    int64
 1   Business Travel     142 non-null    object
 2   Education           142 non-null    object
 3   Gender              142 non-null    object
 4   Age                 142 non-null    int64
 5   MonthlyIncome (K)   142 non-null    float64
 6   Total Working Years 142 non-null    int64
 7   satisfaction_score  142 non-null    int64
 8   Review              140 non-null    object
dtypes: float64(1), int64(4), object(4)
memory usage: 10.1+ KB
```

As shown in the Table 4, the variables, *Business Travel, Education and Gender* are categorical variables, *Review* variable is open-ended text while all other variables are continuous variables. All the categorical variables were provided as objects which will need to be converted to numerical values when conducting the data analysis.

Table 4 Variables of employee_satisfaction_data datafile

Variable Name	Description/Question	Options	Type of variable	Class of variable
Demographic information				
Rowid	Represents a unique identifier for each row in the dataset.		int64	
Business Travel	Indicates if the employee travels away from home for work	• Non-Travel (relabelled = 1) • Travel_Rarely (2) • Travel_Frequently (3)	object	Categorical, ordinal
Education	The level of education of the employee	• A Level (1) • Higher Certificate (2) • Diploma (3) • Bachelors (4) • Masters (5) • Doctorate (6)	object	Categorical, ordinal
Gender	Gender of the employee	• Male • Female	object	Categorical
Age	Age of the employee (years)		int64	Continuous
MonthlyIncome (K)	Monthly income of the employee in thousands.		float64	Continuous
Total Working Years	Total number of years the employee has been working.		int64	Continuous
Satisfaction measures				
Satisfaction Score	Employees were asked to rate their satisfaction with various elements of their work environment, and this score represents the average of those ratings.		int64	Continuous
Review	Open-ended feedback provided by employees about their experience working at ServiceFirst.		object	Text

2.3 Verify data quality and rectify any issues

To assess the quality of the data and rectify issues if required, the following steps were performed:

- Determine if there are any missing data,
- Determine if there are any duplicate data,
- Identify incorrect values by examining unique values of the categorical data,
- Determine if there are any outliers.

Missing values

There were only two missing values in the *Review* column as follows:

	Rowid	Business Travel	Education	Gender	Age	MonthlyIncome (K)	Total Working Years	satisfaction_score	Review
43	44	Travel_Frequently	A Level	Female	39	7.88	18	40	NaN
105	115	Travel_Rarely	A Level	Male	27	5.87	8	72	NaN

For the k-mean analyses and decision tree modelling these rows were not removed as the *Review* column was not utilised. However, they were removed for the sentiment analysis (Section 4.3).

There were no duplicate rows or incorrect values.

Outliers and anomalies in the continuous variables were checked during data exploration by performing descriptive statistics and examining distributions through visualisations (Section 2.5).

2.4 Preprocessing of data

Our analyses for this project will include k-means clustering and decision tree modelling. For k-means clustering, it is required that:

- All data is numerical.
- Integer data type is converted to float data type as k-means clustering relies on distance calculations between data points. These calculations often involve division and square roots, resulting in fractional values.
- Any outliers are minimised or removed as outliers can skew the centroids.
- The data is normalised or standardised if they are measured on different scales.
- The variables are independent of each other.

For decision tree modelling, all categorical data need to be numerical, and the dependent variable will need to converted to a categorical variable if it is a continuous variable.

Therefore our initial preprocessing included converting the variables *Age, Tenure and Satisfaction* from int64 to float64. The conversion of categorical variables to numerical was conducted after the exploration stage to assist with understanding the outputs.

2.5 Exploration report

The exploratory data analysis will provide insights into the characteristics and relationships between variables, which will inform the subsequent k-means clustering and decision tree modelling.

The data was explored as follows:

- Calculate descriptive statistics of continuous and categorical data.
- For continuous data variables, check for outliers, distributions and relationships.

Categorical variables

Bar charts showing the proportion of values for each of the categorical variables, *Gender, Education and Travel* for business are presented in Figures 1 to 3. The majority of employees surveyed were male (62%) and most employees had either an A level education (38%) or Higher certificate (22%). Just over half (52%) did not travel while just over a third travelled frequently (37%). These distributions highlight the diversity within the workforce and suggest potential areas to investigate further, such as the impact of travel frequency and education on satisfaction.

Figure 1 Gender

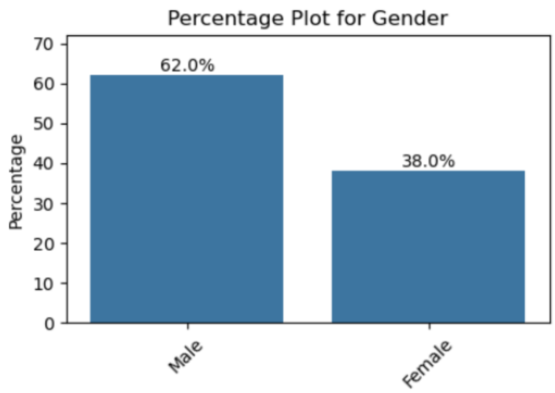


Figure 2 Level of education

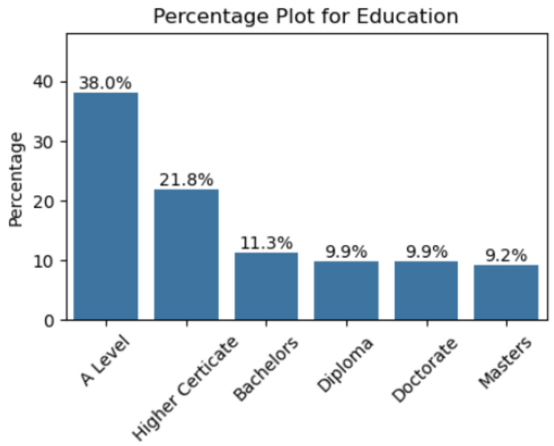
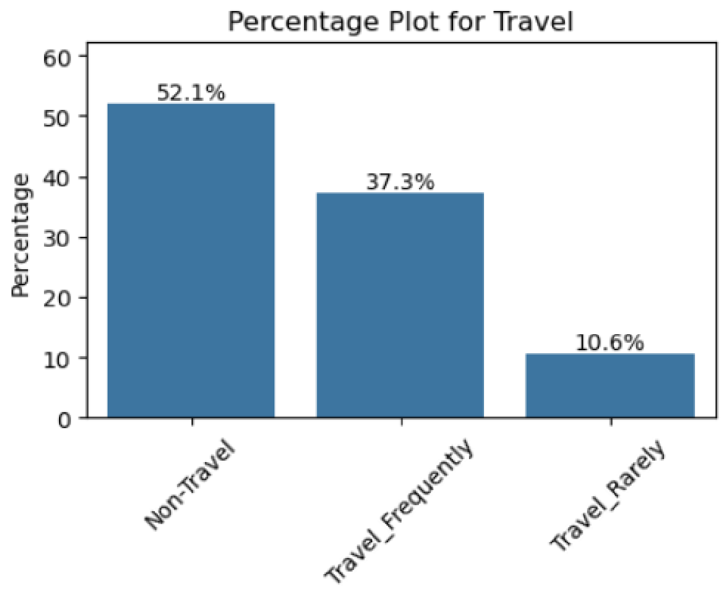


Figure 3 Amount of Travel



Continuous variables

Table 5 provides the mean, standard deviation and interquartile range (IQR) of each of the continuous variables *Age*, *Income*, *Tenure* and *Satisfaction*.

Table 5 Descriptive statistics for Age, Income, Tenure and Satisfaction.

	Age	Income	Tenure	Satisfaction
count	142.000	142.000	142.000	142.000
mean	37.028	8.355	13.444	59.486
std	10.772	5.623	8.945	23.680
min	19.000	1.560	0.000	20.000
25%	27.000	4.070	6.250	37.000
50%	37.000	6.340	10.000	64.000
75%	45.750	14.350	20.750	73.000
max	58.000	20.240	36.000	100.000

The results indicate that:

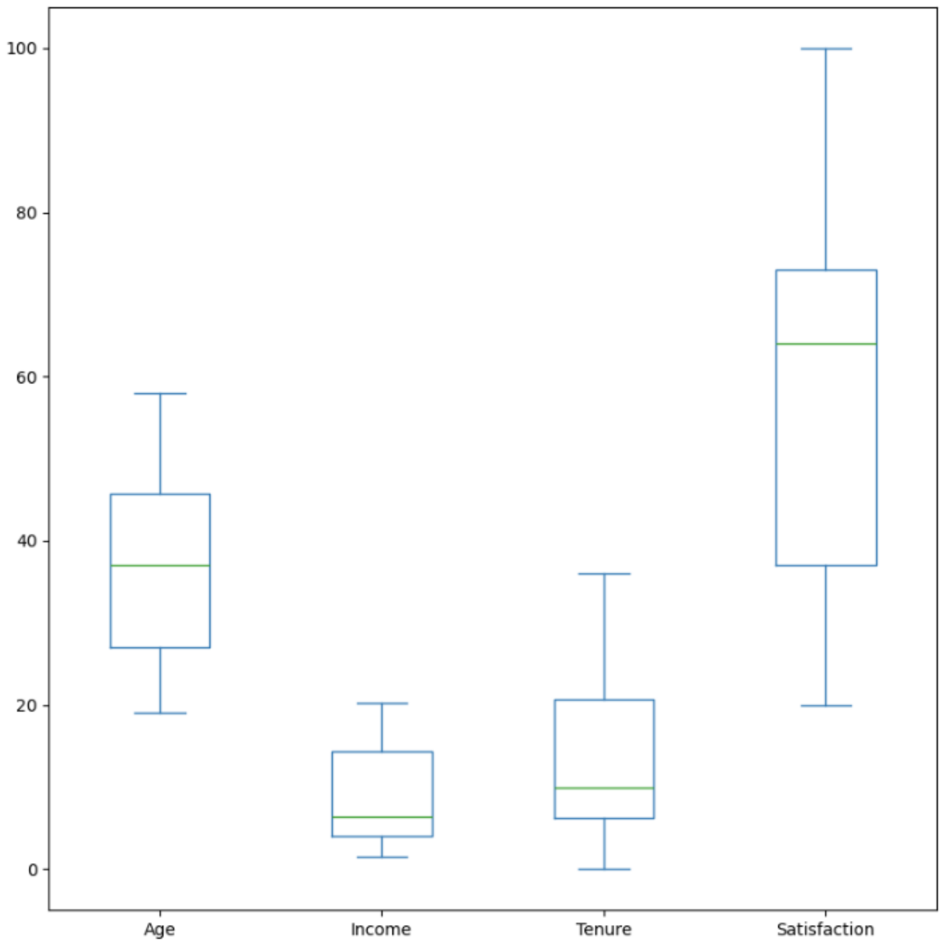
- Age ranged between 19 years to 58 years with an average of 37 years of age.
- Monthly income ranged between $2,000 per month to $14,000 per month. The average was $8,000 per month.
- The total years working (tenure) ranged between 0 years and 36 years with an average of 13 years.
- Satisfaction scores ranged between 20 points to 100 points with an average of 59 points.

For k-means modelling the data was scaled as satisfaction scores have a much greater range than income and tenure (refer to Figure 4).

Outliers

There was no indication of outliers based on the box plots shown in Figure 4.

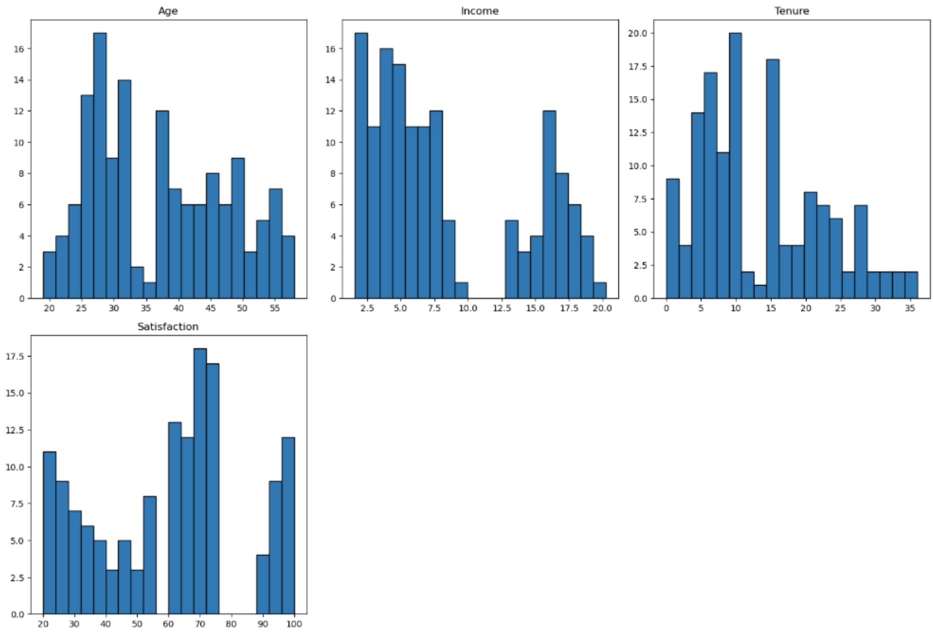
Figure 4 Box plots for variables Age, Income, Tenure and Satisfaction.



Distributions of continuous variables

Figure 5 provides the histograms for the continuous variables *Age, Income, Tenure and Satisfaction* scores. The distributions tend to follow normal distributions except there are some peaks. For *Age*, there seems to be two peaks of normally distributed variables, cut off at around 35 years. For *Income,* there seems to be two peaks of normally distributed variables, those below $10,000 per month and those above $12,000 per month. For *Tenure* there also seems to be two peaks. However for *Satisfaction,* there seems to be three.

Figure 5 Histograms for variables Age, Income, Tenure and Satisfaction scores



Pairplots were also conducted with the categorical variables used as the hue i.e., *Travel*, *Education*, or *Gender* and no hue (Figures 6 to 9).

Figure 6 Pairplot for Age, Income, Tenure and Satisfaction scores (no hue)

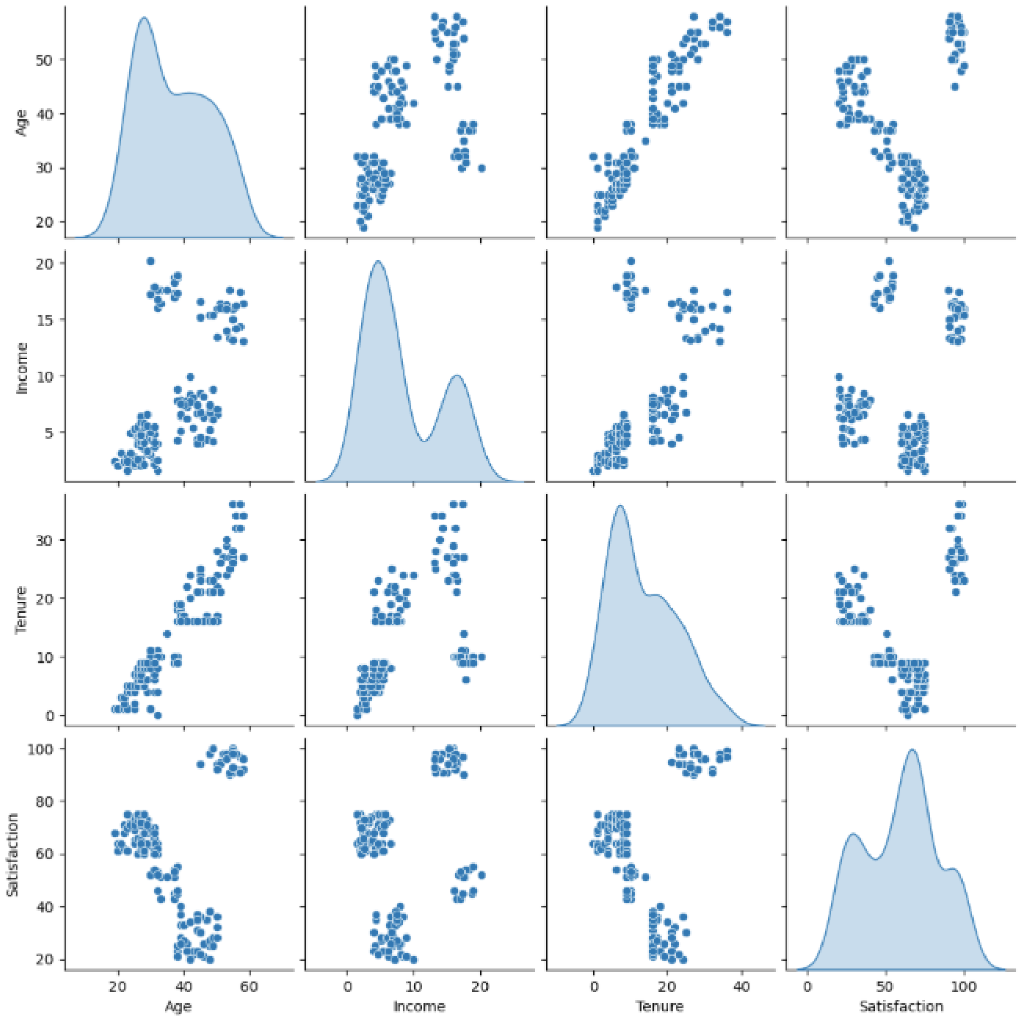


Figure 7 Pairplot for Age, Income, Tenure and Satisfaction scores (hue = Travel)

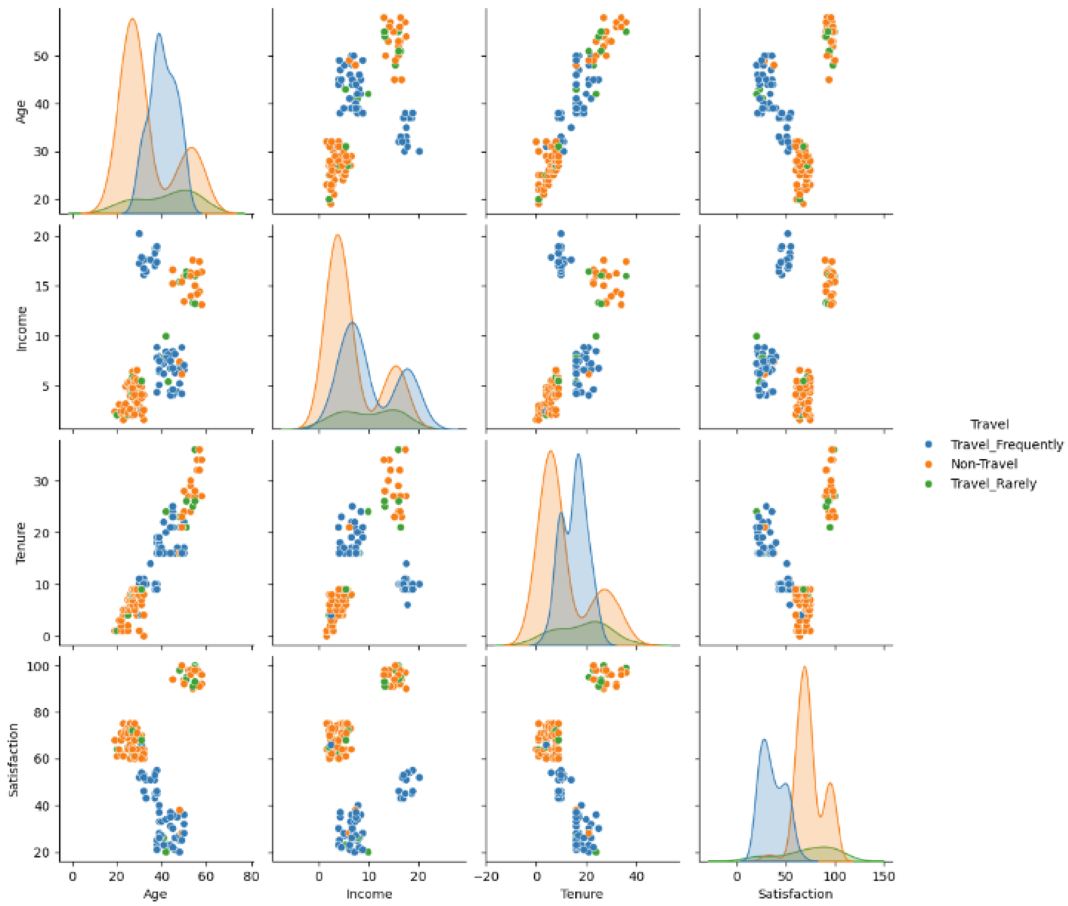


Figure 8 Pairplot for Age, Income, Tenure and Satisfaction scores (hue = Education)

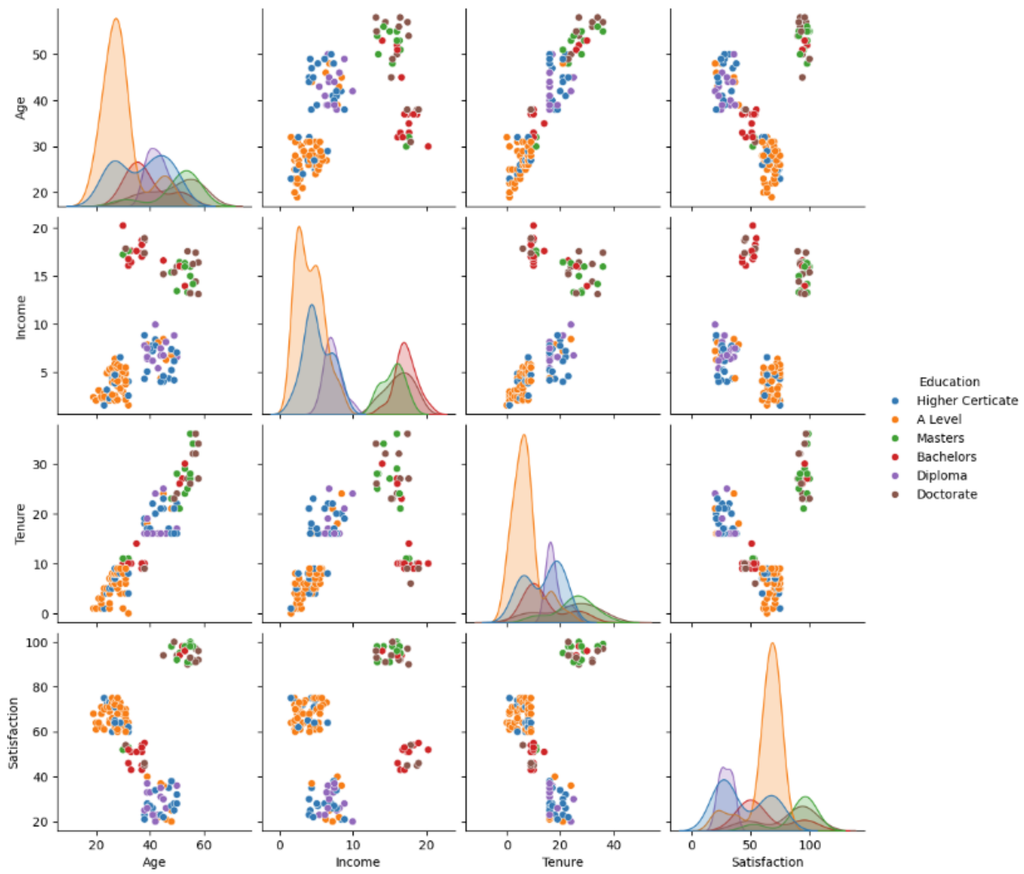
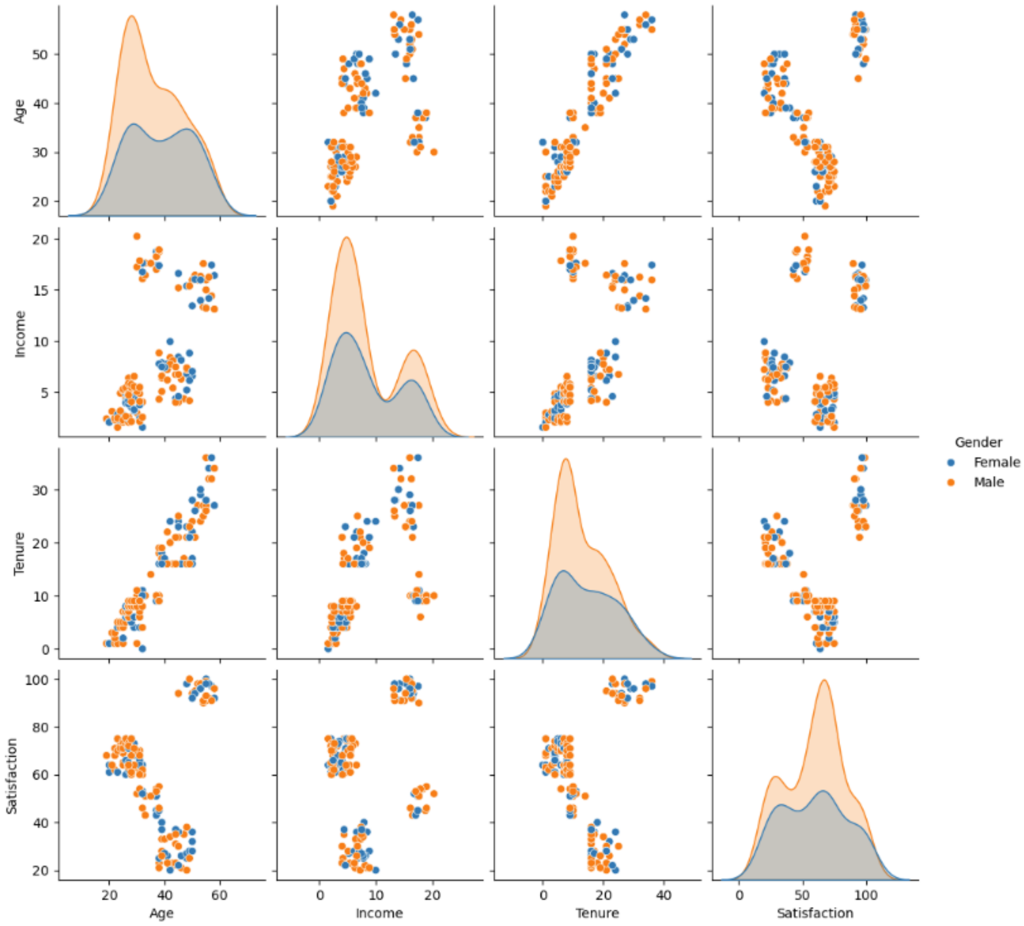


Figure 9 Pairplot for Age, Income, Tenure and Satisfaction scores (hue = Gender)



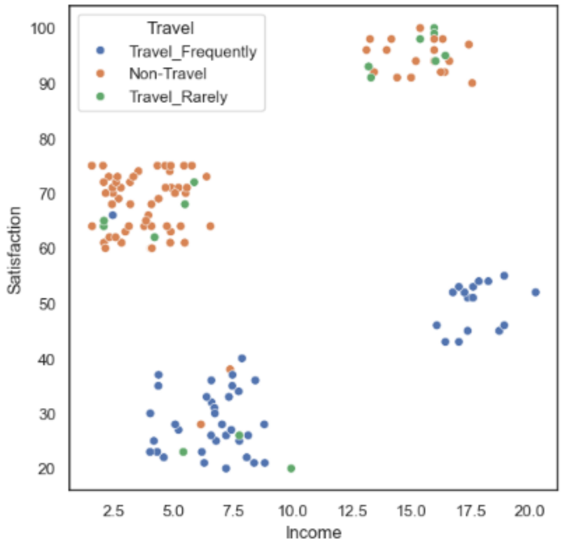
Figures 6 to 9 indicate the following:

- All the continuous variables, *Age, Income, Tenure* and *Satisfaction*, follow a normal distribution. However as confirmed previously in the plots of the histograms (Figure 5), there are multiple peaks in the distributions.
- There seems to be no difference in the distributions or clusters for male or females, however there seems to be differences in the distributions or clusters when comparing education levels and extent of business travel.
- For *Income*, there are two peaks and this was the same regardless of gender and travel frequency. Each peak seems to correspond with education level with those in the lower income range had either A level, Higher Certificate or Diploma, while those in the upper income range had either Bachelors, Masters or Doctorate. This is what would be expected, the higher the education levels the higher the income.
- For *Age* and *Tenure* there are two peaks for those that do not travel, whereas those that travel frequently or not at all had only one peak.
- There seems to be a linear relationship between *Tenure* and *Age* i.e., the longer the tenure the older the employee.
- Four clusters are shown when viewing the *Satisfaction* vs *Income* chart with different distributions for *Education* and *Travel.*
- There seems to be two clusters for *Satisfaction* vs *Age* and for *Satisfaction* vs *Tenure*.

The histograms and pairplots confirm normal distribution trends, albeit with multiple peaks for *Age, Income, Tenure* and *Satisfaction.* This multimodality suggests the existence of distinct subgroups within the dataset, a hypothesis supported by pairplots, which revealed potential clustering patterns.

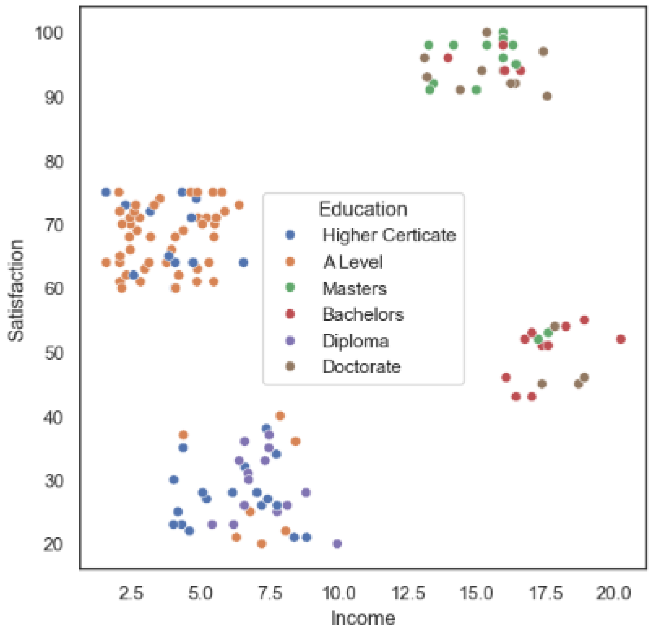
To further understand the relationship between *Satisfaction* and *Income,* scatterplots were created, with hue=Travel and hue=Education (Figures 10 and 11, respectively). Figure 10 indicates that the top two clusters have more people who are non-travellers compared to the bottom two clusters who seem to travel frequently. Employees that travelled less frequently or not at all were more satisfied that those that travelled frequently, with this observation being the same whether in the lower or higher income bracket.

Figure 10 Scatter plot of Satisfaction vs Income (hue = Travel)



When reviewing the scatterplot for hue=Education (refer Figure 11) the top right cluster seems to be employees with mainly a Masters level or Doctorate, top left seems to be mainly A level, bottom left is a mixture of mainly Higher Certificate or Diploma, while bottom right is mainly Bachelor. Even so, the separation of the clusters based on *Education* are not as distinct as compared to *Travel.*

Figure 11 Scatter plot of Satisfaction vs Income (hue = Education)



Correlations between variables

Further exploration of the data was conducted by determining the correlation between the variables using a heatmap. The correlations between all the variables (including categorical, once converted to numerical values), were investigated.

Figure 12, provides a pairplot to visualise these relationships. The heatmap in Figure 13, displays the correlation coefficients (Pearson's r), quantifying the strength and direction of the relationship between pairs of variables in the regression analysis. The following was found:

- *Satisfaction* has a very weak to almost no linear correlation with the variables A*ge* (r = 0.13), *Income* (r = 0.22) and *Tenure* (r = 0.11). This confirms what was viewed in the pairplots (Figure 6 to 9)
- There is an extremely strong positive correlation between *Tenure* and *Age* (r = 0.93), that is the longer they have been with the company, the older the employee.
- There is a medium to substantial positive correlation between *Income* and *Age* (r = 0.57) and *Income* and *Tenure* (r = 0.55).
- Of the categorical variables:
    - As expected being a non-ordinal categorical variable, there is no linear relationship between *Gender* and any other variable.
    - However, as *Travel* and *Education* are ordinal categorical variables, correlations with the continuous variables were explored:
        - *Travel* has a very weak positive linear relationship with education (r = 0.19).
        - *Travel* has a weak to moderate positive linear relationship with *Tenure*, *Income* and *Age* (0.21≤ r≤ 0.31).
        - There is a strong negative relationship between *Travel* and *Satisfaction* (r = -0.73), indicating that frequent travel is associated with lower satisfaction.
        - *Education* has a weak to moderate positive relationship with S*atisfaction* (r = 0.31). That is the higher the *Education*, the higher the *Satisfaction*
        - *Education* has a medium to substantial positive relationship with *Tenure* and *Age* (0.64≤r≤0.60)
        - *Education* has a very strong positive relationship with *Income* (r = 0.89). That is the higher the *Education* the higher the *Income*.

Figure 12 Pairplot for all variables (hue = Travel)

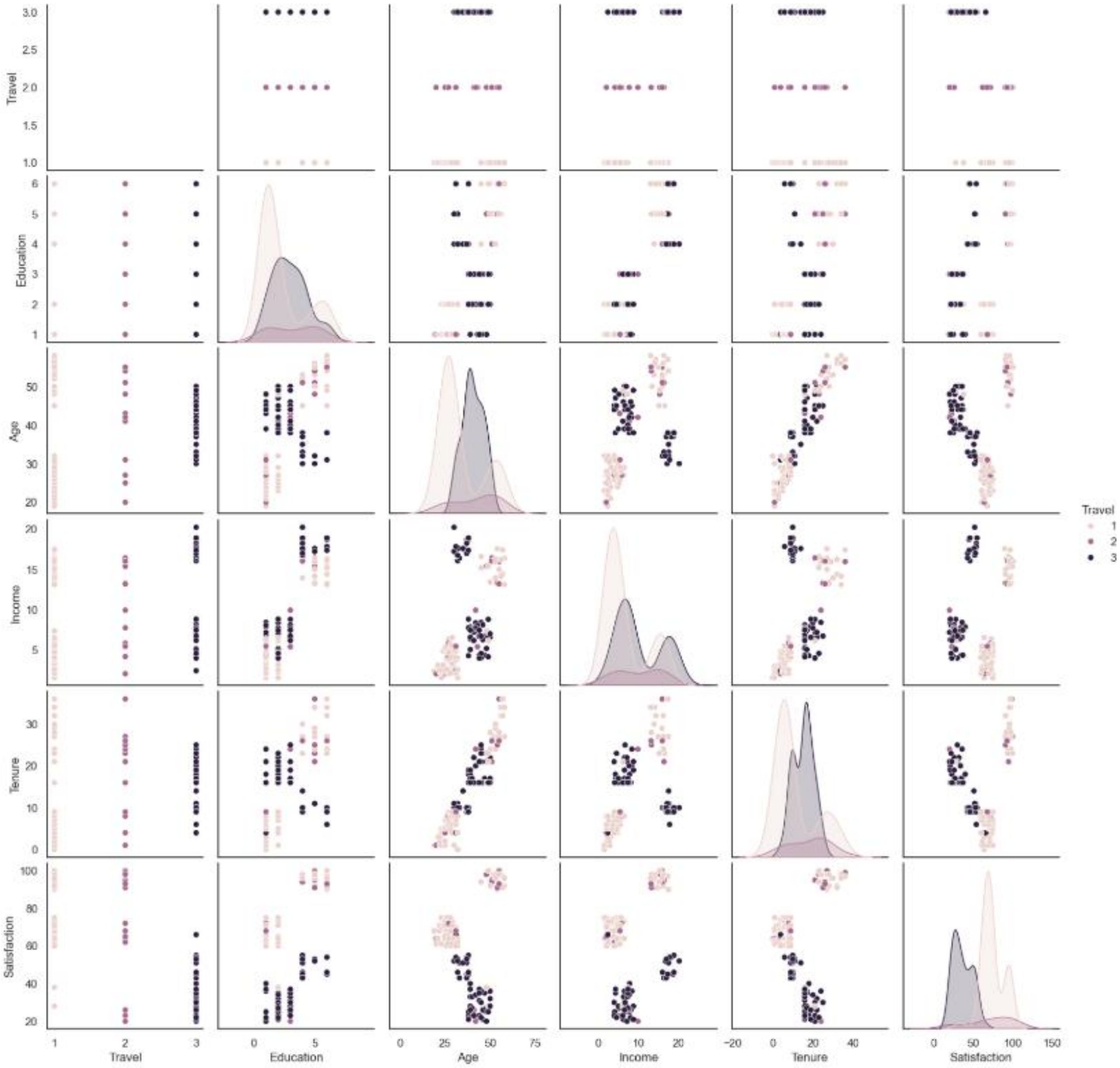
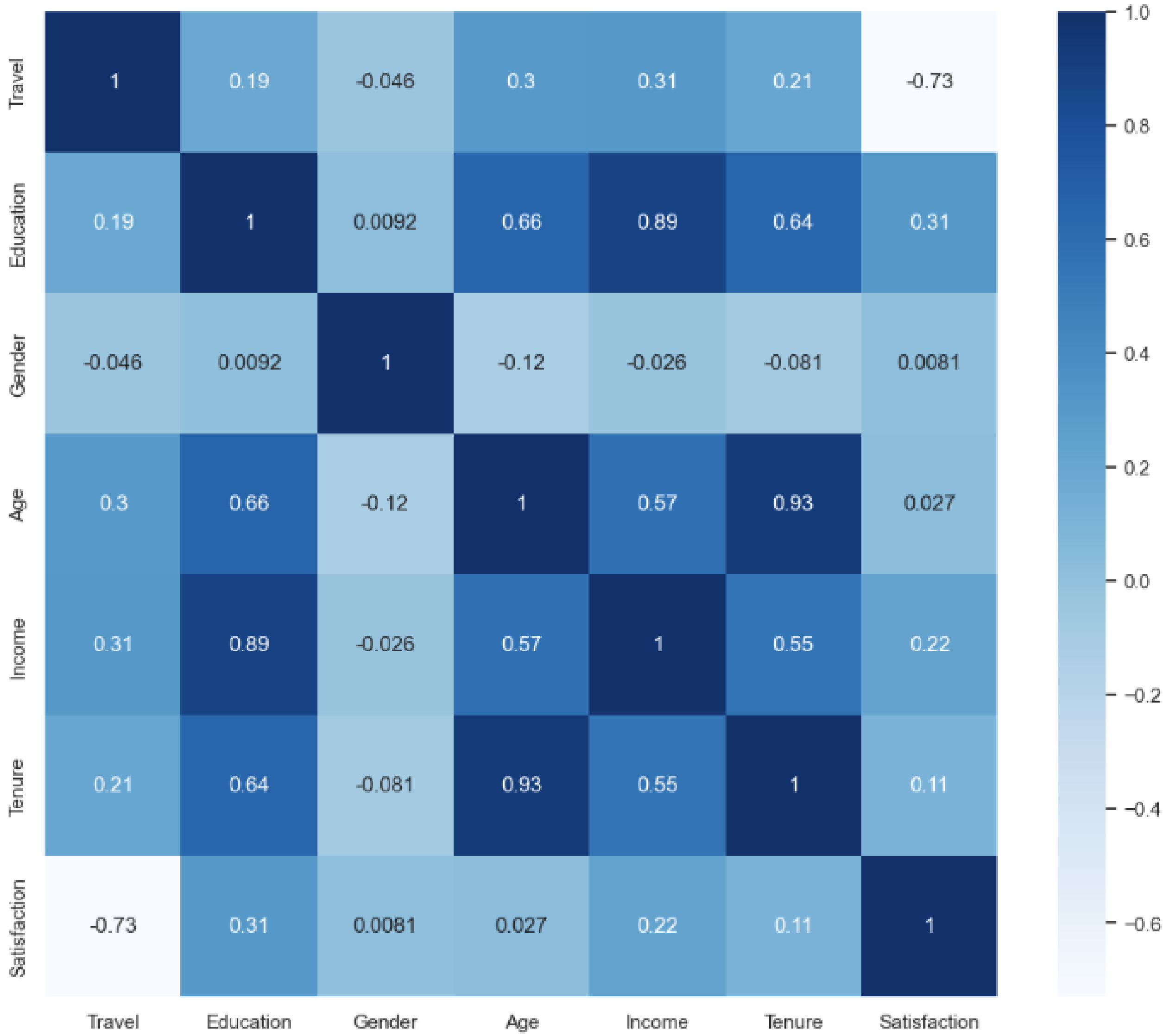


Figure 13 Heatmap of correlations between variables



The exploratory analysis has provided a strong foundation for the next steps:

- **K-means clustering**: The observed clustering patterns in *Satisfaction* and *Income*, along with the presence of multimodal distributions, justify the application of k-means to identify distinct employee groups.
- **Decision tree modelling**: The weak linear relationships between *Satisfaction* and other variables highlight the need for decision tree modelling to uncover non-linear interactions and identify key predictors of *Satisfaction.*

3 Data preparation

The selection, construction and reformatting of the data for modelling is described in Section 4.

However, in summary the dataframes within Jupyter notebook were labelled as follows:

- survey = full data set provided (*employee_satisfaction_data.csv* ') (142 rows, 9 columns)
- survey_c = includes only continuous variables (*Age, Income, Tenure, Satisfaction*) used during the exploration of the continuous variables.
- survey_r = includes all variables except *Rowid and Review*
- survey_d = includes all variables except *Rowid, Review* and *Cluster*
- survey_s = includes only *Review* and *Cluster* variables.

3.1 Clean data

Please refer to Section 2.3 and 2.4.

4 Modelling and testing

4.1 K-means clustering

K-means clustering modelling was utilised to address the first business goal of identifying groups of employees with similar satisfaction levels and monthly incomes. K-means clustering is a machine learning algorithm that groups data points into a specified number of clusters by minimising the distance between points and their cluster centres. It assumes that clusters are roughly spherical, variables contribute equally and distances are meaningful, with results influenced by initial centre placement and the chosen number of clusters.

a. Prediction

Preliminary data exploration suggests that there are four potential clusters explaining the relationship between satisfaction scores and monthly income. However, to validate the optimal number of clusters, the Elbow method and Silhouette method was employed.

b. Data preparation

The data was first standardised as *Satisfaction* scores had a significantly higher range compared to *Income*, *Tenure* and *Age*. Standardisation ensures that all features have a mean of 0 and a standard deviation of 1, which is critical for clustering algorithms to perform effectively. Normalisation was not required as the data was confirmed to follow a normal distribution.

This was achieved with the **StandardScaler** from **sklearn.preprocessing**. The data was scaled using the **fit_transform method**. The output in Figure 14, confirms that the variables were standardised as all variables had a mean of 0 and a standard deviation of 1. The scaled dataframe was called, **X_scaled_df.**

Figure 14 Output of standardised variables

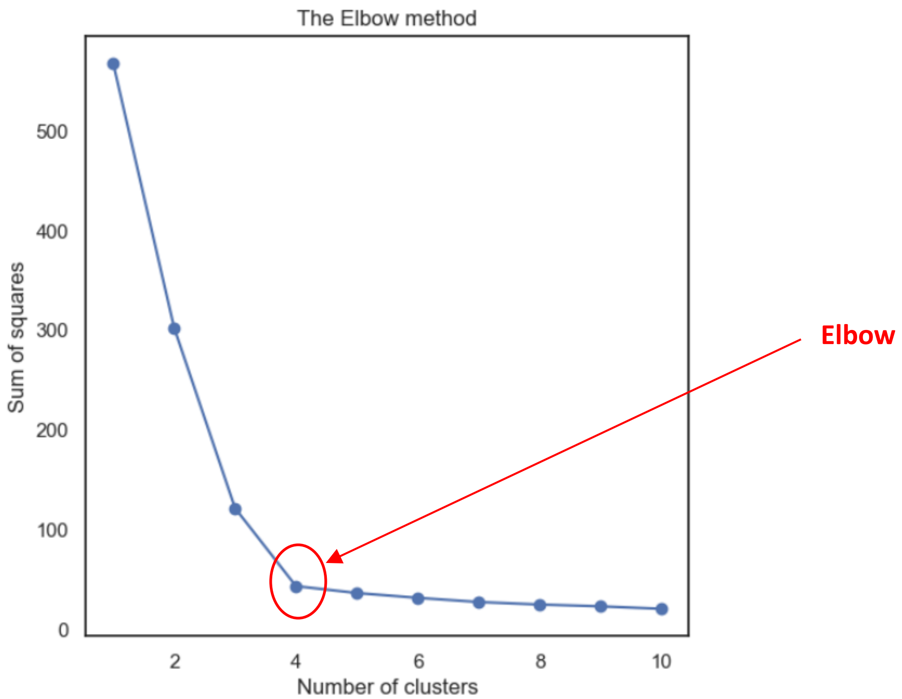
	Age	Income	Tenure	Satisfaction
count	142.00	142.00	142.00	142.00
mean	0.00	-0.00	0.00	0.00
std	1.00	1.00	1.00	1.00
min	-1.68	-1.21	-1.51	-1.67
25%	-0.93	-0.76	-0.81	-0.95
50%	-0.00	-0.36	-0.39	0.19
75%	0.81	1.07	0.82	0.57
max	1.95	2.12	2.53	1.72

c. Determine optimal number of clusters using both the Elbow Method and the Silhouette Method.

Elbow Method

The Elbow method helps identify the optimal number of clusters in k-means by plotting the within-cluster sum of squared errors (SSE) against the number of clusters. The "elbow point" on the graph indicates where adding more clusters yields diminishing returns in reducing SSE. This is achieved by iteratively fitting **KMeans** with varying cluster numbers, storing the SSE values, and plotting them. Using **KMeans** from **sklearn.cluster**, parameters like **n_clusters** and **init='k-means++'** ensure efficient clustering. The scaled data **X_scaled_df** is used. The plot to determine the optimal clusters is provided in Figure 15.

Figure 15 Plot of sum of squares against number of clusters (k) for the Elbow method



As shown in Figure 15, the SSE reduces as the number of clusters increases. The elbow occurs at k=4 where the change of SSE diminishes significantly as the number of clusters increases.

Silhouette Method

The Silhouette method evaluates the quality of clustering by measuring how similar data points are to their own cluster (cohesion) compared to other clusters (separation). It calculates a Silhouette score (Sil) for each data point, ranging from -1 to 1, where higher scores indicate better-defined clusters. The average Silhouette score across all data points helps determine the optimal number of clusters. This was achieved by using **silhouette_score** from **sklearn.metrics**, which computes the score for different cluster counts, allowing the selection of clusters with the highest average score. The plot of scores (Sil) against number of clusters is provided in Figure 16.

Figure 16 Plot of Silhouette score (Sil) against number of clusters

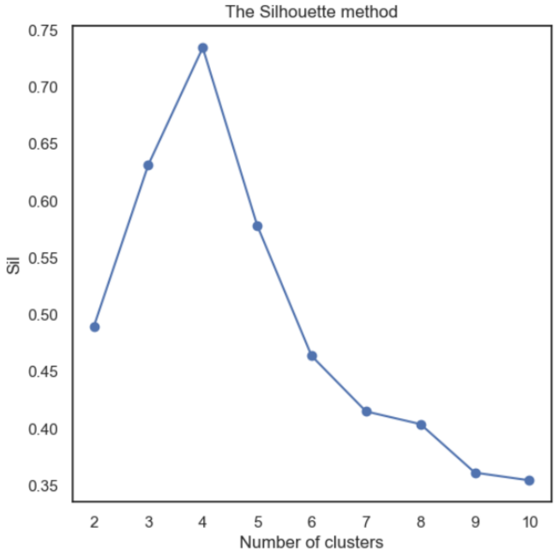


Figure 16 indicates that the highest average Sil score corresponds to a k-value of 4, followed by k=3 then k=5. Based on both the Elbow method and the Silhouette method, the optimal number of clusters seems to be 4. However, the clusters for both k=3 and k=4 were evaluated and visualised to determine which clustering produced the most meaningful and useful results.

d. *Evaluation and visualisation of clusters*

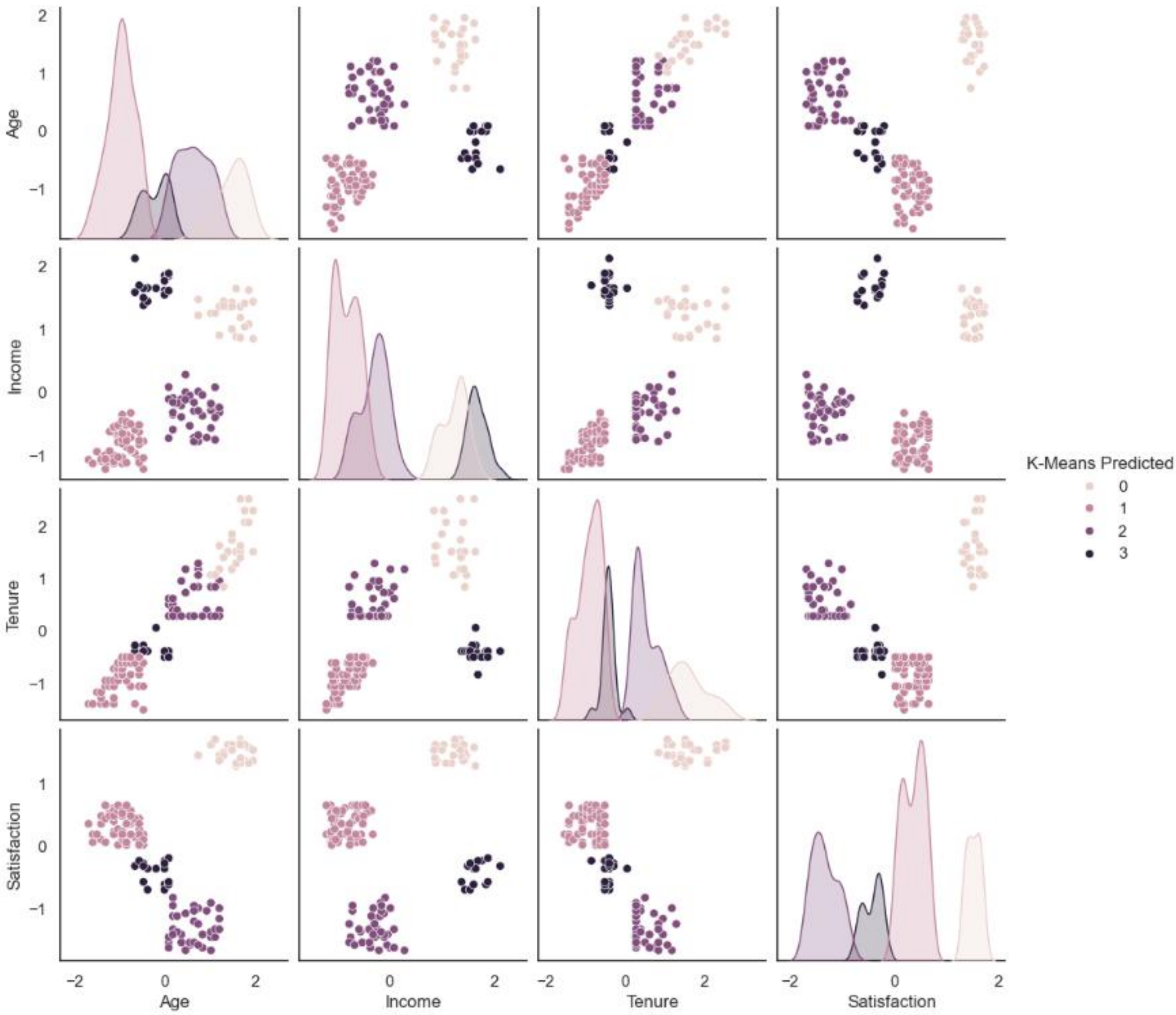
The methods used to evaluate and visualise the clusters for either k=3 or k=4 were as follows:

1. Producing visualisations via pairplots,
2. Calculating the v-measure,
3. Calculating and comparing the means of the variables *Satisfaction*, *Income*, Age and *Tenure* for each cluster
4. Calculating and comparing the frequency of values for the categorical values *Travel* and *Education*
5. Calculating and comparing the cluster cardinality and magnitude for each cluster

Evaluation of cluster when k=4

1. ***Pairplot.*** The pairplot between *Satisfaction*, *Income*, *Age* and *Tenure* is provided in Figure 17. The scatterplots indicated four distinct clusters across the pairings, *Satisfaction* with *Age*, *Income*, and *Tenure*, as well as *Income* with *Tenure* and *Income* with *Age*. These clusters were well-defined and non-overlapping, indicating hard clustering.

Figure 17 Pairplot for Satisfaction, Income, Age and Tenure (hue = k-means predicted)



2. ***V-measure.*** The V-measure measures the balance between homogeneity (how similar the elements within each cluster are) and completeness (how well all elements of a given true class are assigned to the same cluster). Its value ranges between 0 and 1, the closer to 1 the higher the agreement between the predicted clusters and the true labels. The v-measure was calculated using **v_measure_score** from **sklearn.metrics.** The true labels utilised were *Education* and *Travel* and the v-measures were as follows:

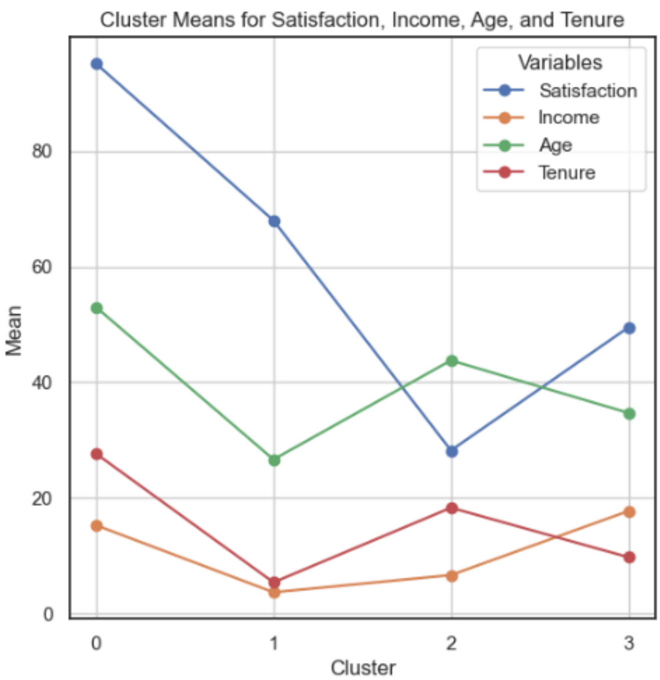
- Education v-measure = 0.57. Indicates a moderate agreement between clusters and the true labels of Education. The clustering captured some patterns that align with education.
- Travel v-measure = 0.498. Indicates some level of agreement between clusters and the true labels of Travel. However, this agreement was weaker compared to the V-measure for *Education*.

3. Means of Satisfaction, Income, Age and Tenure for each cluster. A table of means and a line plot is provided in Table 6 and Figure 18. *Satisfaction* scores vary significantly among clusters, with Cluster 0 having the highest average *Satisfaction* (95.08) and Cluster 2 the lowest (28.23). Conversely, Cluster 0 also has a relatively low average *Income* (15.31), while Cluster 3, with a moderate *Satisfaction* score (49.50), has the highest *Income* (17.76). Thus there doesn't seem to be a clear pattern in the relationship between *Satisfaction* and *Income* levels. Other attributes like *Age* and *Tenure* also show variation between *Satisfaction* scores without a clear pattern. .

Table 6 Means of Satisfaction, Income, Age and Tenure for each cluster

Cluster	Satisfaction	Income	Age	Tenure
0	95.08	15.31	53.00	27.68
1	67.97	3.72	26.67	5.43
2	28.23	6.69	43.79	18.33
3	49.50	17.76	34.72	9.78

Figure 18 Lineplot of means of Satisfaction, Income, Age and Tenure for each cluster



4. *Frequency of Education and Travel for each cluster.* A table of frequency of Travel (1=Non-Travel; 2=Travel_Rarely; 3=Travel_Frequently) and Education (1=A Level; 2=Higher Certificate; 3=Diploma; 4=Bachelors; 5=Masters; 6=Doctorate) for each cluster is provided in Table 7.

Table 7 Frequency of Travel and Education for each cluster

K-Means Predicted	0	1	2	3
Travel				
1	18	54	2	0
2	7	5	3	0
3	0	1	34	18
K-Means Predicted	0	1	2	3
Education				
1	0	47	7	0
2	0	13	18	0
3	0	0	14	0
4	4	0	0	12
5	11	0	0	2
6	10	0	0	4

For *Travel*, Cluster 0 and 1, predominantly includes individuals who travel rarely or not at all, while Clusters 2 and 3 is mostly associated with frequent travellers.

For *Education*, Cluster 1 and 2 primarily consists of individuals with A-Level qualifications, Higher Certificates and Diplomas. However, Cluster 3 is mostly associated with individuals holding Bachelor's degrees, while Clusters 0 show higher representation among those with advanced qualifications like Masters and Doctorates. These patterns highlight how *Travel* and *Education* categories help distinguish the clusters.

4. Cluster cardinality and magnitude Cluster cardinality involves determining the number of data points in each cluster. If one cluster has significantly high cardinality, it indicates the k-means clustering algorithm may have over-generalised and has grouped too many diverse data points together. A small cluster may indicate the presence of outliers. Cluster magnitude measures the sum of distances from the centroid values to determine anomalies. Clusters with unusually high magnitudes may indicate anomalies. It should be noted that clusters with higher cardinality usually have higher magnitudes due to the increased number of points contributing to the sum. Plotting these two metrics together can reveal patterns or outliers. A scatterplot with a best-fit line helps identify deviations. Cluster cardinality, magnitudes and relationship between the two are provided in Figures 19 to 21.

Figure 19 Cluster cardinality, k=4

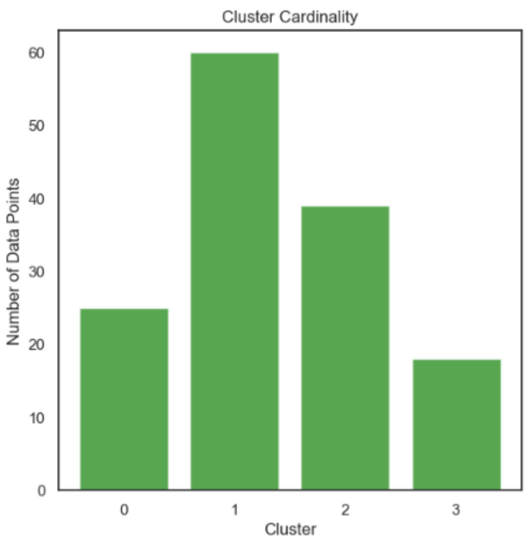


Figure 20 Cluster magnitude, k=4

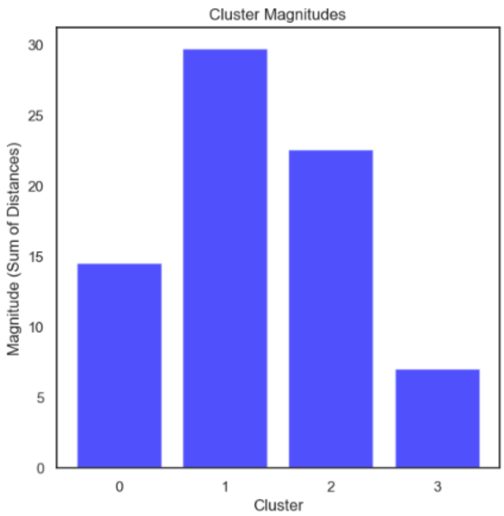


Figure 21 Relationship between cluster cardinality and magnitude

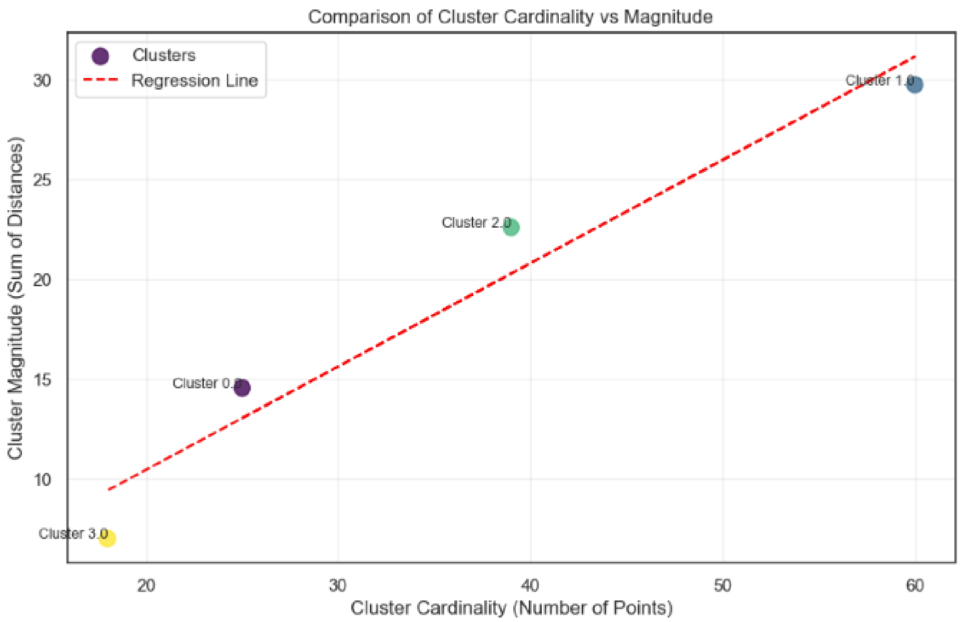
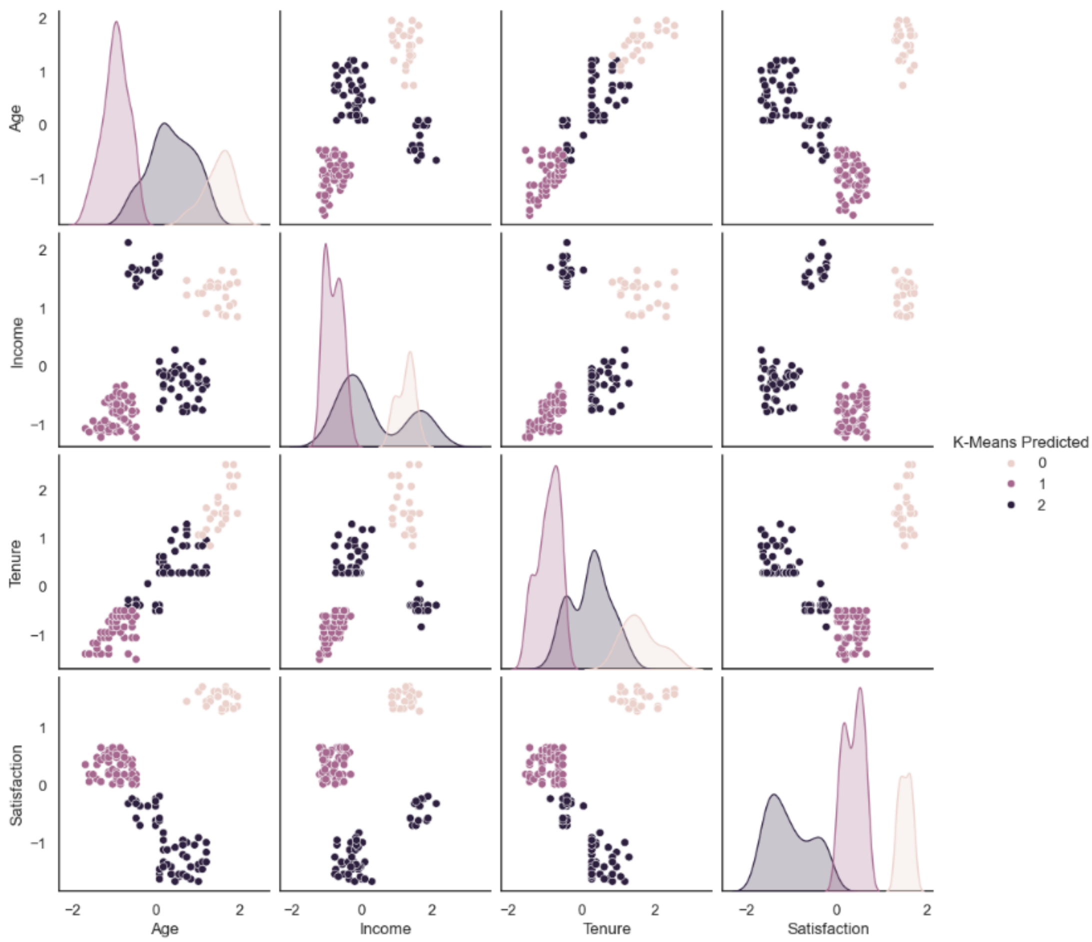


Figure 19 indicates that the there is an uneven distribution of data points with Cluster 1 the largest and Cluster 3 the smallest. A similar relationship was found with cluster magnitude (distance between centroids). Figure 21 indicates a positive relationship between cluster size (cardinality) and spread (magnitude). The larger cluster (Cluster 1), is less compact, while the smallest cluster, Cluster 3, is tighter and more cohesive. The clusters above the regression line (Cluster 0 and 2) may lack compactness, which suggests opportunities to improve the clustering by balancing size and cohesion.

Evaluation of cluster when k=3

1. Pairplot. The pairplot between *Satisfaction*, *Income*, *Age* and *Tenure* is provided in Figure 22. Compared to when k=4, the k=3 clustering seems to have combined clusters the 2 and 3 together. The clustering for *Satisfaction* vs *Age* and *Satisfaction* vs *Tenure* is better for when k=3 compared to when k=4.

Figure 22 Pairplot for Satisfaction, Income, Age and Tenure (hue = k-means predicted)



2. V-measure. The v-measures for *Education* and *Travel* were as follows:

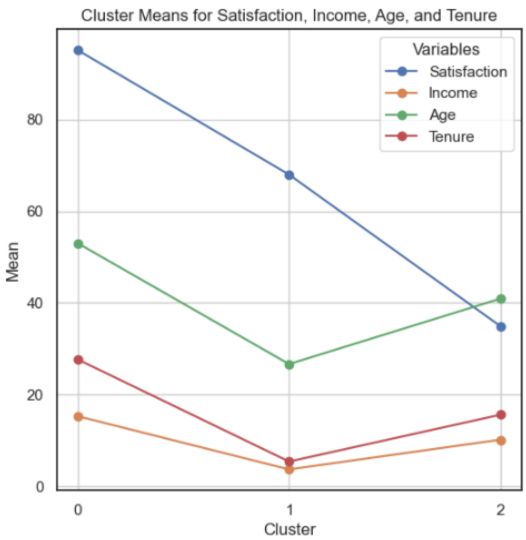
- Education v-measure = 0.44. Although there is some level of agreement, the v-measure for education is lower for k=3 then when k=4.
- Travel v-measure = 0.54. Indicates a moderate level of agreement between clusters and the true labels of *Travel* and this time stronger than for *Education.* It was also stronger than for when k=4.

3. Means of Satisfaction, Income, Age and Tenure for each cluster. A table of means and a line plot is provided in Table 8 and Figure 23.

Table 8 Means of Satisfaction, Income, Age and Tenure for each cluster

Cluster	Satisfaction	Income	Age	Tenure
0	95.08	15.31	53.00	27.68
1	67.97	3.72	26.67	5.43
2	34.95	10.19	40.93	15.63

Figure 23 Lineplot of means of Satisfaction, Income, Age and Tenure for each cluster



The outputs indicates a decline in *Satisfaction* scores as clusters transition from Cluster 0 to Cluster 2. The highest *Satisfaction* scores (Cluster 0) was associated with the highest *Income, Age* and *Tenure* (**High Group**). The middle *Satisfaction* Score (Cluster 1) was associated with the lowest *Income, Age* and *Tenure* (**Low Group**), while the lowest *Satisfaction* score (Cluster 2) was associated with the middle *Income, Age and Tenure* means (**Medium Group**). The k=3 clustering highlights a more structured segmentation, with clear distinctions in *Income*, *Satisfaction*, *Age*, and *Tenure* suggesting that this configuration may better capture meaningful differences across the data.

4. Frequency of Education and Travel for each cluster. A table of frequency of Travel (1=Non-Travel; 2=Travel_Rarely; 3=Travel_Frequently) and Education (1=A Level; 2=Higher Certificate; 3=Diploma; 4=Bachelors; 5=Masters; 6=Doctorate) for each cluster is provided in Table 9.

Table 9 Frequency of Travel and Education for each cluster

K-Means Predicted	0	1	2
Travel			
1	18	54	2
2	7	5	3
3	0	1	52
K-Means Predicted	0	1	2
Education			
1	0	47	7
2	0	13	18
3	0	0	14
4	4	0	12
5	11	0	2
6	10	0	4

Again for *Travel*, Cluster 0 and 1, predominantly includes individuals who travel rarely or not at all, while Clusters 2 is mostly associated with frequent travellers.

For *Education*, Cluster 0 show higher representation among those with advanced qualifications like Masters and Doctorates. Cluster 2 primarily consists of individuals with A-Level qualifications and Higher Certificates. However, Cluster 3 seems to include all education levels which a greater representation of mid-level education, Higher Certificates, Diplomas and Bachelor degrees.

4. Cluster cardinality and magnitude Cluster cardinality, magnitudes and relationship between the two are provided in Figures 24 to 26

Figure 24 Cluster cardinality, k=3

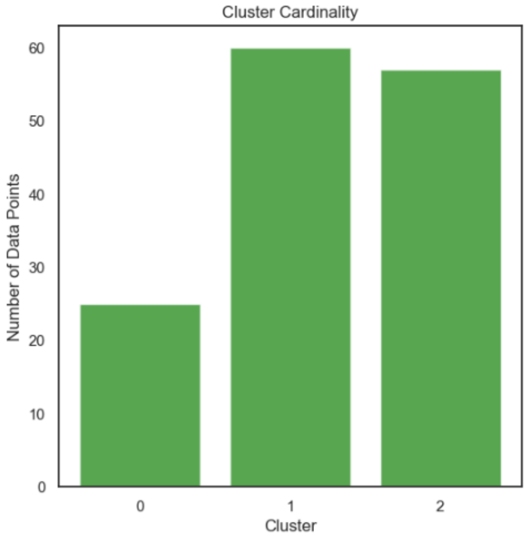


Figure 25 Cluster magnitude, k=3

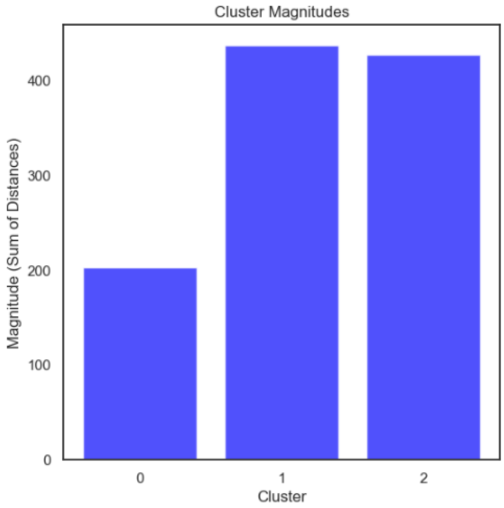


Figure 26 Relationship between cluster cardinality and magnitude

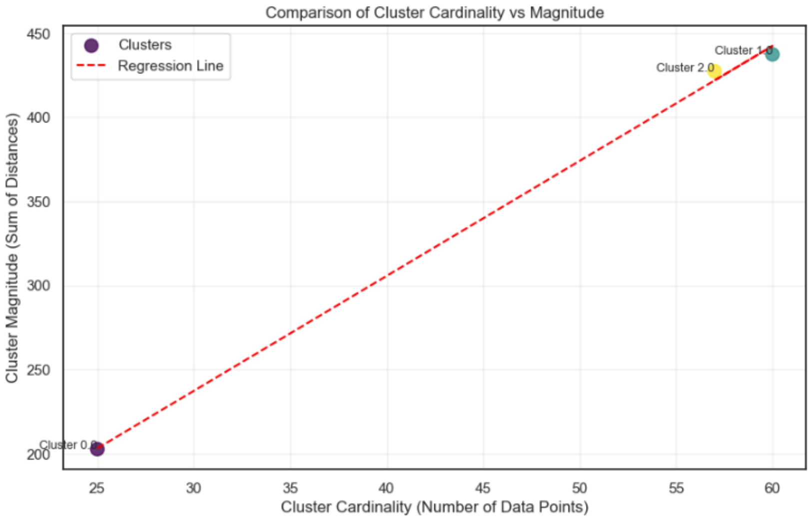


Figure 24 indicates that the there is still an uneven distribution of data points. However, Clusters 1 and 2 have similar number of data points while Cluster 0 has the least. A similar relationship was found with cluster magnitude (distance between centroids). Figure 26 indicates a positive relationship between cluster size (cardinality) and spread (magnitude). The larger clusters (Cluster 1 and 2), are less compact, while the smallest cluster, Cluster 0, is tighter and more cohesive.

Discussion of evaluation

Based on the evaluation, the choice between whether to have 3 or 4 clusters depended on the specific priorities for the clustering analysis.

For k=4, the clustering created four distinct, non-overlapping clusters in the pairplot. The v-measures for *Education* (0.57) and *Travel* (0.498) suggest moderate alignment with true labels, with better agreement for *Education* compared to *Travel*. However, the clusters showed an uneven distribution, with Cluster 1 being the largest and least compact, while Cluster 3 was the smallest and most cohesive. There was no clear pattern between *Satisfaction* and *Income*, and the clusters showed some lack of compactness, indicating potential improvement opportunities.

For k=3, the clustering merges two clusters (Cluster 2 & 3) from the k=4 scenario, resulting in a more structured segmentation. The pairplots show better clustering for *Satisfaction* vs. *Age* and *Satisfaction* vs. *Tenure*, and the outputs indicate a clearer decline in *Satisfaction* scores as clusters transition from Cluster 0 (highest *Satisfaction*, *Income*, *Age* and *Tenure*) to Cluster 2 (lowest *satisfaction* but moderate *Income*, *Age*, and *Tenure*). The v-measures show a trade-off: *Travel* improves to 0.54, while *Education* declines to 0.44, suggesting a shift in alignment toward travel-related clustering. Additionally, the clusters are more balanced in size, with Clusters 1 and 2 being larger and less compact, while Cluster 0 is smaller and more cohesive.

The k=3 clustering appears to offer more meaningful segmentation due to its clearer patterns, better-balanced cluster sizes, and well-defined relationships among *Satisfaction*, *Income*, *Age* and *Tenure*. A notable insight from this clustering is the relationship between *Travel* and *Satisfaction*. Employees who travelled most frequently were the least satisfied, while those who travelled rarely or not at all were more satisfied. This pattern reveals distinct groups within the workforce.

Cluster 0 consists of employees who rarely or never travel and had the highest *Satisfaction* scores. These individuals are characterised by either having the highest income, advanced education levels (e.g., Masters or Doctorates), are older and have been with the organisation the longest (**High Group**). These attributes suggest that this group may include senior employees, those in management positions or research positions, who are less likely to be required to travel due to the nature of their roles.

Cluster 1 also composed of employees who rarely or never travel, stands in contrast to the **High Group**. These employees are characterised by the lowest income, lower levels of education (e.g., A-Levels or Higher Certificates), younger age and shorter tenures (**Low Group**). This group likely includes newer employees who are still early in their careers and may not yet have roles that require or allow frequent travel. This group had the second highest *Satisfaction* levels.

Cluster 2 primarily includes frequent travellers. This group exhibits middle levels of *Income*, *Age*, *Tenure*, and *Education* (**Medium Group**) and is the least satisfied overall. Frequent travel may contribute to dissatisfaction due to work-life imbalance, stress and in turn reduced job satisfaction (Ivancevich et al., 2003). This group may include employees in mid-level roles or operational positions where travel is a job requirement but without the benefits or stability of senior roles.

These findings suggest a potential link between employee satisfaction, travel requirements and career progression. Employees in senior roles may avoid frequent travel, while mid-level employees bear the burden of travel demands, contributing to lower satisfaction. In contrast, entry-level employees, despite travelling less, might experience lower satisfaction due to limited compensation, shorter tenure and fewer opportunities.

4.2 Decision tree modelling

So far, k-means modelling has indicated a relationship between *Satisfaction* levels and *Travel* moderated by *Income, Age, Tenure* and *Education* . To further understand this relationship and address our second business goal, decision tree modelling was utilised to predict satisfaction levels based on various factors.

Decision tree modelling is a machine learning technique that splits data into branches based on feature values to make predictions or classify outcomes. Each split is chosen to maximise the separation of data points based on the target variable, creating a tree structure where leaves represent final predictions or classifications.

It assumes that relationships between features and the target variable can be captured through hierarchical, rule-based splits. The method is sensitive to data quality, as noisy or unbalanced data can lead to overfitting. It also assumes that splits are meaningful and that sufficient data is available to support the tree's depth and complexity.

a. Prediction

Based on the exploration of the data as well as the k-means modelling, *Satisfaction* levels may be predicted based on *Travel*, *Tenure*, *Age* and *Education*.

Although *Gender* did not seem to have a relationship with Satisfaction levels, it was included in the initial classification modelling.

b. Data preparation

Decision tree modelling classifies data into different categories or classes. Thus to use the decision model to predict satisfaction levels (dependent variable, i.e., y), the *Satisfaction* variable was converted from a continuous variable to a categorical variable (*Satisfaction_overall)*, as follows:

- Satisfied = Satisfaction scores >=50
- Dissatisfied = Satisfaction scores <50

c. Decision tree modelling

The decision modelling involved:

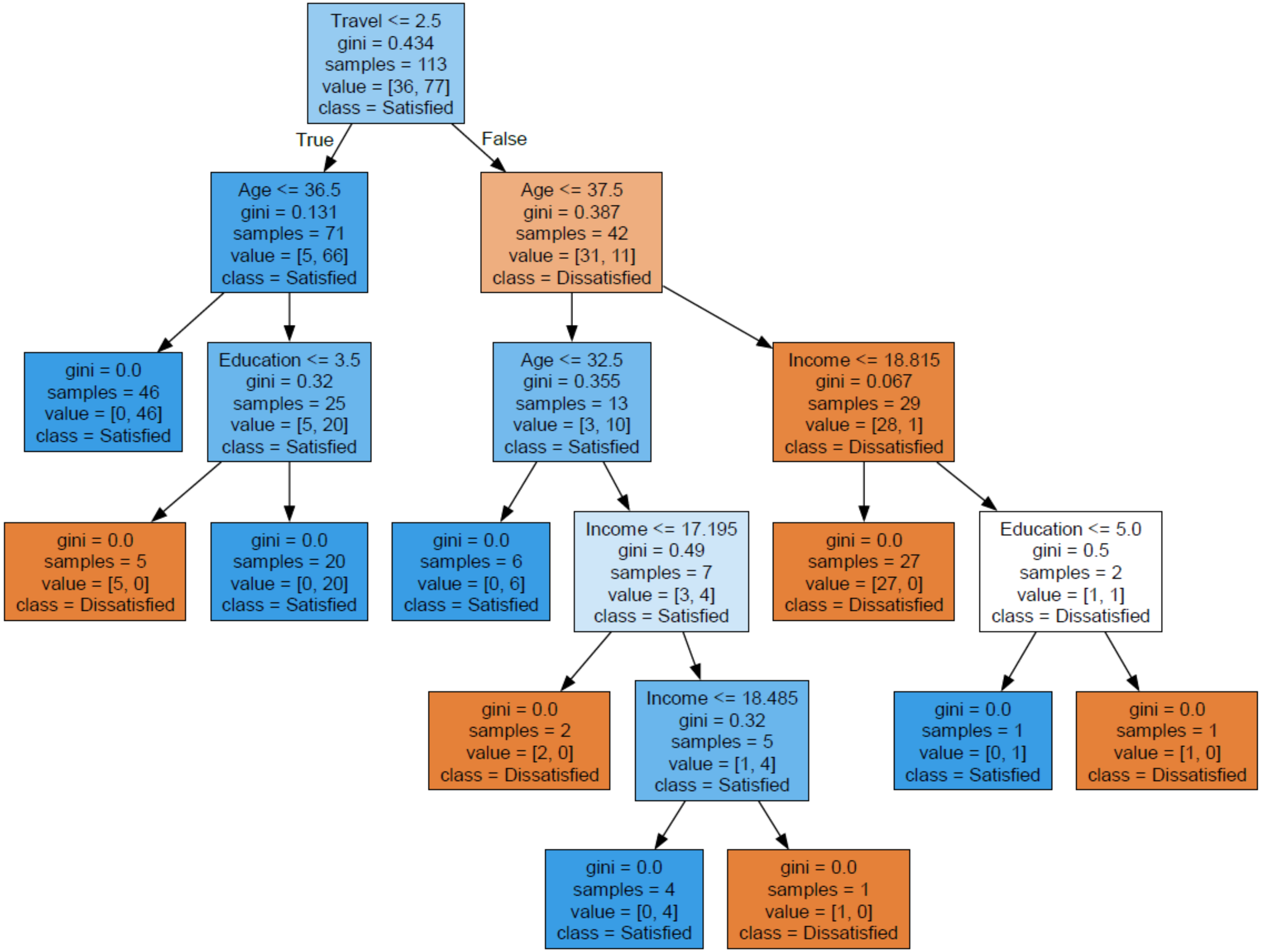
- Setting the variable (X (predictors) = *Travel*, *Education*, *Gender*, *Age*, *Income* and *Tenure* and y (dependent variable) = *Satisfaction_overall*)
- Splitting the data into training (80%) and testing (20%) using **train_test_split** from **sklearn. model_selection** library
- Specifying and fitting the model using **DecisionTreeClassifier** from **sklearn.tree** library**.** The criterion for the model was the Gini method. The Gini value represents the Gini impurity, which measures how "pure" or homogeneous a node is in terms of its classification. Gini ranges from 0 to 1, with the higher the value the more diversity at the node.
- Testing the model by determining the predicted values of y based on the X_test data set with the **predict() method**.
- Visualising the model via a Textual tree (using **tree** and **export_text** from **sklean** library) and Graphviz tree (using **graphviz** and **export_graphviz** from **sklearn.tree**; **display** from **IPython.display**).

Figures 27 and 28 provides the visualisations of the decision modelling.

Figure 27 Textual tree

```
|--- Travel <= 2.50
|   |--- Age <= 36.50
|   |   |--- class: Satisfied
|   |--- Age >  36.50
|   |   |--- Education <= 3.50
|   |   |   |--- class: Dissatisfied
|   |   |--- Education >  3.50
|   |   |   |--- class: Satisfied
|--- Travel >  2.50
|   |--- Age <= 37.50
|   |   |--- Age <= 32.50
|   |   |   |--- class: Satisfied
|   |   |--- Age >  32.50
|   |   |   |--- Income <= 17.19
|   |   |   |   |--- class: Dissatisfied
|   |   |   |--- Income >  17.19
|   |   |   |   |--- Income <= 18.48
|   |   |   |   |   |--- class: Satisfied
|   |   |   |   |--- Income >  18.48
|   |   |   |   |   |--- class: Dissatisfied
|   |--- Age >  37.50
|   |   |--- Income <= 18.81
|   |   |   |--- class: Dissatisfied
|   |   |--- Income >  18.81
|   |   |   |--- Education <= 5.00
|   |   |   |   |--- class: Satisfied
|   |   |   |--- Education >  5.00
|   |   |   |   |--- class: Dissatisfied
```

Figure 28 Graphviz decision tree



The output of the visuals indicates that the most important feature that predicts *Satisfaction* levels is *Travel,* the first split being those that travel frequently (Travel = 3, travel frequently) are dissatisfied compared to those that travel rarely or not at all (Travel ≤ 2.5, i.e., no travel (1) and travel rarely (2)).

For those who travel frequently, the next predictor was *Age* then *Education*. For those who travel rarely or not at all the next predictor was *Age*, followed by *Income* then *Education*. Neither *Tenure* or *Gender* were predictors in the model. It is noted that the sample numbers in the last few leaves are quite small. Therefore it was considered to prune the tree, but not before evaluating it.

d. Evaluate the decision tree

The accuracy of a model was evaluated by comparing its predictions to the actual values in the testing data set. The **accuracy score**, **confusion matrix** and **classification report** from the **sklearn** library was used to evaluate the model. The output is provided in Table 10.

Table 10 Accuracy score, confusion matrix and classification report

```
Accuracy: 0.9310344827586207

Confusion Matrix:
 [[ 9  1]
 [ 1 18]]

Classification Report:
               precision    recall  f1-score   support

Dissatisfied       0.90      0.90      0.90        10
   Satisfied       0.95      0.95      0.95        19

    accuracy                           0.93        29
   macro avg       0.92      0.92      0.92        29
weighted avg       0.93      0.93      0.93        29
```

The evaluation output provides the following with regard to the predictors *Travel, Age, Income and Education* on satisfaction levels:

- The model is 93.1% accurate overall in correctly predicting those that are dissatisfied and those that are satisfied
- The model correctly predicted 9 cases as dissatisfied (True Negative, TN) and 18 cases as satisfied (True Positive, TP)
- One case was incorrectly predicted as satisfied (False Positive, FP) and one case was incorrectly predicted as dissatisfied (False Negative, FN)

Precision:

- Out of those that have been predicted as dissatisfied (9+1=10; TN+FN) the model correctly predicted 9 cases (90%)
- Out of those that have been predicted as satisfied (18+1=19; TP+FP) the model correctly predicted 18 cases (95%)

Recall:

- Out of all those that were actually dissatisfied (10; TN+FP) the model correctly predicted 9 cases (90%)
- Out of all those that were actually satisfied (19; FN+TP) the model correctly predicted 18 cases (95%)

The F1 Score combines Precision and Recall

- The F1 score for dissatisfaction (0.9) is a little lower than for satisfaction (0.95), thus the model was better at predicting those that are satisfied than those that are dissatisfied.

For ServiceFirst, the priority is likely to accurately identify dissatisfied employees, enabling timely interventions to enhance their job satisfaction and increase retention. To achieve this, further refinement of the decision tree model was undertaken by pruning it to a maximum depth of three levels, ensuring a balance between simplicity and predictive accuracy.

e. Pruning the tree

The pruning process for the decision tree followed the same methodology as previously, with the additional restriction of limiting the tree to a maximum depth of three levels.

As shown in Figure 29, the most important feature that predicts *Satisfaction* remains the amount of travel. By pruning the true, *Education* is no longer a predictor. Instead, *Tenure* is a predictor for those who don't travel or travel rarely (i.e. either Travel = 0 or 1), with *Income* as the next important factor. For employees who travel frequently (i.e., Travel = 3), *Age* emerged as the next predictor, followed by *Income.*

Overall, the pruned decision tree indicates that out of those who travel infrequently or not at all, those with an income less than or equal to $11,535 per month and a *Tenure* of less than or equal to 12.5 years were dissatisfied. This dissatisfaction, appears to be driven primarily by low income levels. It may also reflect a preference among some employees for roles involving travel, which are generally associated with higher remuneration.

For employees who travel frequently, dissatisfaction is associated with incomes of $18,815 per month or less. This indicates that when travel is a significant aspect of a role, adequate compensation becomes crucial to reducing dissatisfaction. This finding suggests that ServiceFirst must ensure employees required to travel are remunerated appropriately, as insufficient compensation may exacerbate dissatisfaction with their role.

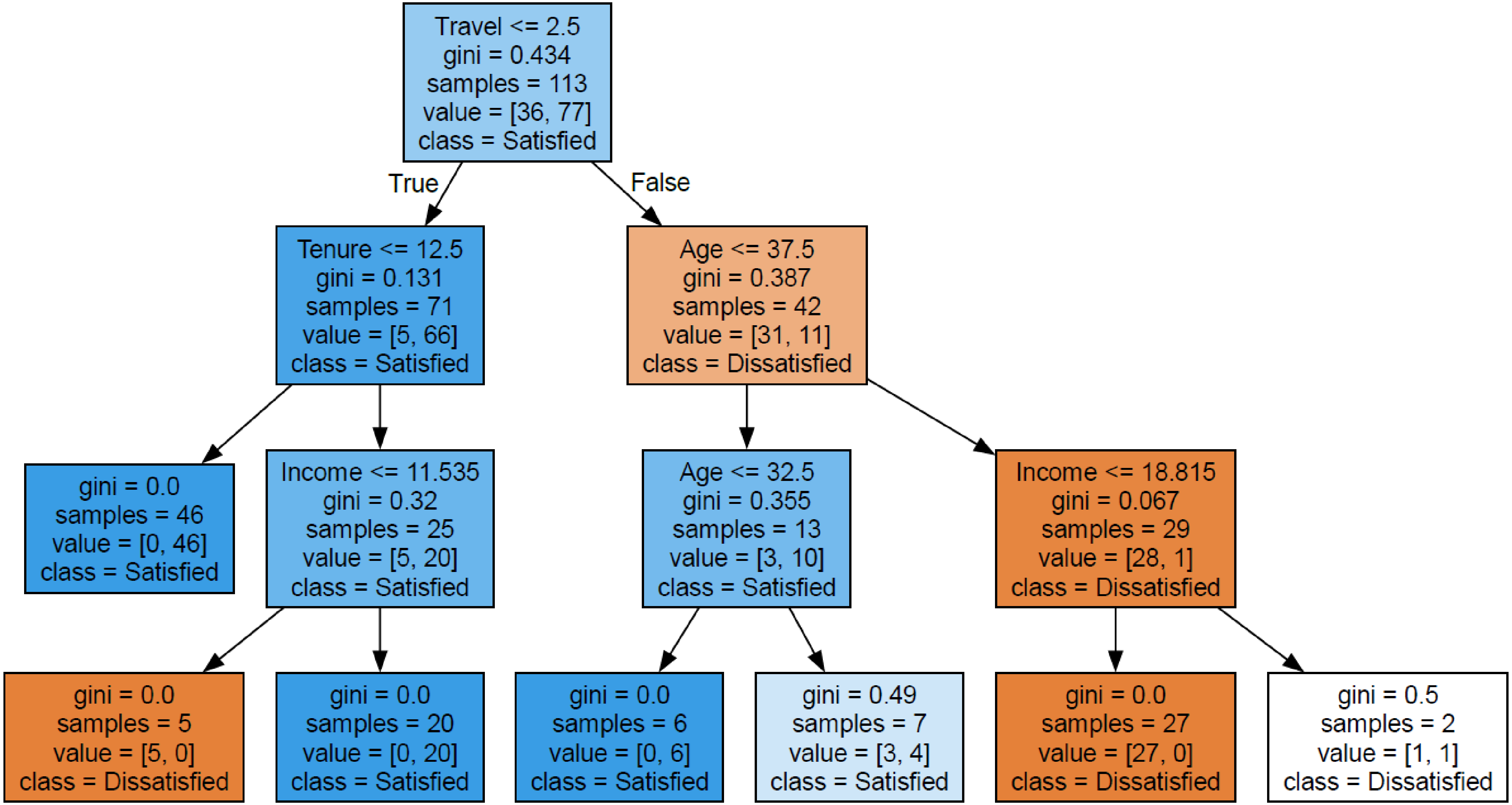
d. Evaluation of the pruned decision tree

The output is provided in Table 11.

Table 11 Accuracy score, confusion matrix and classification report

```
Accuracy: 0.9655172413793104

Confusion Matrix:
 [[ 9  1]
 [ 0 19]]

Classification Report:
              precision    recall  f1-score   support

Dissatisfied       1.00      0.90      0.95        10
   Satisfied       0.95      1.00      0.97        19

    accuracy                           0.97        29
   macro avg       0.97      0.95      0.96        29
weighted avg       0.97      0.97      0.97        29
```

Figure 29 Graphviz pruned decision tree



Even though *Education* is no longer a predictor, by pruning the true, the accuracy of the model has improved to 96.5% with the predictors *Travel, Tenure, Age* and *Income.* The output indicates the following:

- The model correctly predicted 9 cases as dissatisfied (True Negatives)
- The model correctly predicted 19 cases as satisfied (True Positives)
- One case was incorrectly predicted as satisfied (False Positive)
- No case was incorrectly predicted as dissatisfied (False Negative)

Precision:

- Out of those that have been predicted as dissatisfied (9+0=9) the model correctly predicted all cases (100%)
- Out of those that have been predicted as satisfied (19+1=20) the model correctly predicted 19 cases (95%)

Recall:

- Out of all those that were actually dissatisfied (10), the model correctly predicted 9 cases (90%)
- Out of all those that were actually satisfied (19), the model correctly predicted 19 cases (100%)

The F1 Score

- The F1 score for dissatisfaction (0.95) is a still a little lower than for satisfaction (0.97), however it has improved from the previous model which was 0.9.

Overall, the accuracy of the pruned decision tree has increased, and has improved its ability to identify dissatisfied employees while balancing false positives and false negatives more effectively. The improvement in the F1 score for dissatisfaction suggests that the model is now better at correctly recognising and classifying dissatisfied employees, reducing the likelihood of misclassifications and improving its usefulness for targeted interventions. However, there is still a minor disparity in performance compared to satisfaction, indicating potential room for further improvement.

4.3 Sentiment analysis

A sentiment analysis of open-ended feedback was conducted to address the third business goal. This involved:

- Preprocessing and tokenising employee feedback.
- Creating visualisations and determine frequency distributions and
- Reviewing the polarity and sentiment of the feedback to identify the top positive and negative sentiments.

a) Data preparation

Data preparation involved retaining only the *Review* column (which contains the written feedback) and the *Cluster* column. The *Cluster* column represents the k-means predicted groupings (0, 1, and 2) when k=3. These groupings were used to compare sentiment scores for the *Rowid* entries. Two missing values in the *Review* column were deleted.

b) Data preprocessing

The data was preprocessed to ensure the feedback was in the same format for ease of analysis and comparison. The following preprocessing actions were conducted:

- Ensuring data was in string format (converting *Review* column datatype from **object** to **string**)
- Changing all the words to lowercase (using the **lambda** function)
- Expanding any contractions (using **contractions** function)
- Removing punctuations (via **replace** function)
- Removing duplicates

Duplicates

In the original dataframe *survey* it was found that there were no duplicates. However, the preprocessed dataframe found that for the *Review* column there were ten duplicates. A review of the duplicates found that most of the actual comments, e.g., *'no comment', 'it is okay' and 'love sf'*, are common phrases. However, the phrase *'servicefirst is a good place to start'* and *'servicefirst is a great start for my career learning a lot'* is too unique, so only the duplicates of these statements were removed. The final dataframe had 138 rows of data and three columns, with all comments in *Review column* in lower case, no punctuations and expanded contractions (refer to Figure 30).

Figure 30 Review column output (top 20)

```
stuck with low wages for years management does not seem to care
servicefirst is a great start for my career learning a lot
i am learning a ton here love it
servicefirst is a great start for my career learning a lot
i have been with servicefirst for over a decade and the ability to work from home has been a gamechanger for my worklife balance
i feel like i never see my kids because i am always traveling for work
pays not high but the experience is worth it
getting good experience here
loving the learning opportunities at servicefirst
been here forever but the pay is still terrible too much travel too
great place to gain experience
gaining valuable skills
wish pay was higher it is too low
way too much travel for such low pay
servicefirst is giving me great experience
it is good
low income  it is not making me happy
too much travel wish i could be home more often
servicefirst offers excellent benefits and a fixed schedule which allows me to plan my personal life better go sf
not a fan of the ceo
```

c) Conduct visualisations and determine frequency distributions of most common words

To conduct visualisations and determine frequency distributions of the most common words the following steps were performed first:

- Tokenisation to split all the words into single words (using **punk_tab** package and **word_tokenize** function from **nltk** library)
- Normalisation using lemmatization (converts words to their dictionary form (e.g., "swim," "swam," "swum" → "swim"). It is much more accurate and context-aware than stemming (using **WordNetLemmatizer**, **wordnet**, **pos_tag** from **nltk** library)
- Elimination of stop words (e.g., and, was, a etc) (using **stopwords** from **nltk** library)

Lemmatisation is the process of reducing words to their base or root form, known as a "lemma," while considering the word's context and meaning. Unlike stemming, which trims words to their root by removing suffixes, lemmatisation uses linguistic rules and dictionaries to ensure the base form is a valid word (e.g., "running" becomes "run," and "better" becomes "good").

Prior to lemmatisation it was ensured that the grammatical role of the word was tagged (i.e., verb, noun, adjective etc) before being lemmatised. It ensured a more accurate and context-sensitive processing of words by considering the grammatical role of the text. The output is provided in Figure 31, confirming lemmatisation, e.g., ‘stuck’ has been converted to ‘stick, ‘does’ to ‘do’, ‘is’ to ‘be’ etc.

Figure 31 Lemmatisation output

```
Original Tokens: ['stuck', 'with', 'low', 'wages', 'for', 'years', 'management', 'does', 'not', 'seem', 'to', 'care', 'servicefirst', 'is', 'a', 'great', 'start', 'for', 'my', 'career']
POS Tags: [('stuck', 'VBN'), ('with', 'IN'), ('low', 'JJ'), ('wages', 'NNS'), ('for', 'IN'), ('years', 'NNS'), ('management', 'NN'), ('does', 'VBZ'), ('not', 'RB'), ('seem', 'VB'), ('to', 'TO'), ('care', 'VB'), ('servicefirst', 'JJ'), ('is', 'VBZ'), ('a', 'DT'), ('great', 'JJ'), ('start', 'NN'), ('for', 'IN'), ('my', 'PRP$'), ('career', 'NN')]
Lemmatized Tokens: ['stick', 'with', 'low', 'wage', 'for', 'year', 'management', 'do', 'not', 'seem', 'to', 'care', 'servicefirst', 'be', 'a', 'great', 'start', 'for', 'my', 'career']
```

The removal of stopwords reduced the number of words in our Token list from 1143 to 595 separate words.

The distributions of the words were visualised via word cloud and frequency distributions. The initial visualisations found that words, ‘work’, ‘good’, ‘servicefirst’, ‘pay’ and ‘sf’, were the top five most frequent words. As servicefirst and sf (abbreviation of servicefirst) is the name of the company, these were removed to produce the wordcloud and frequency distribution shown in Figure 32 and 33 respectively. The final Token list had 564 words with the top five most frequent words being ‘work’, ‘good’, ‘pay’, ‘travel’ and ‘low’.

Figure 32 Wordcloud

Figure 33 Barplot of top 30 most frequent words

d) Determine sentiment (positive, neutral, negative) of comments

Two methods were used to determine the sentiment or emotional tone of the comments (positive, neutral, negative).

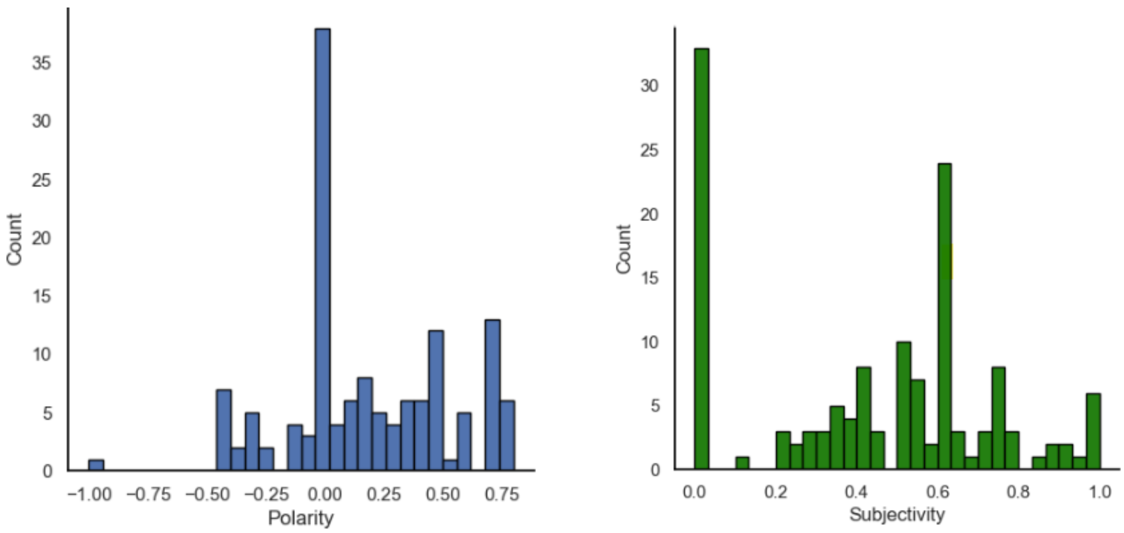
The first method that was initially utilised was the **Textblob** sentiment analysis tool from the **textblob** library. **Textblob** measures both polarity and subjectivity. The polarity measure provides the sentiment or emotional tone of the text, indicating how positive, negative or neutral. The value ranges from -1 (most negative) to +1 (most positive). A value of 0 is neutral. The subjectivity measure provides how subjective or objective a statement (i.e., whether it is an opinion or fact). The value ranges from 0 (completely objective) to + 1 (completely subjective). However, **Textblob** has been found to struggle with negative sentiments (Fesenko, 2020).

Therefore, **NLTK Vader** sentiment analysis tool was also utilised which has been found to be better at predicting negative sentiments and better suited for informal text. It provides more nuanced sentiment intensity scores (Brown, 2024). **Vader** from the **nltk** library categorises text as either positive, negative or neutral to then determine a weighted summation score called ‘compound’ which ranges from -1 (highly negative) to 1 (highly positive).

Textblob method

The histograms of the polarity score (sentiment) and subjectivity score are provided in Figure 34.

Figure 34 Histograms of polarity and subjectivity score



For polarity, the highest frequency of the comments were scored as neutral. There were a few negative comments otherwise most were positive. The mean polarity was found to be 0.18, which was just above neutral. It was also found that 25% of the comments were above the polarity score of 0.5 (Table 12).

Table 12 Descriptive statistics for polarity

```
count    138.00
mean       0.18
std        0.35
min       -1.00
25%        0.00
50%        0.12
75%        0.50
max        0.80
Name: Polarity, dtype: float64
```

For subjectivity, the highest frequency of the comments have been scored as zero, i.e., they were objective comments. However, the mean subjectivity score was found to be 0.4, indicating most of the comments are opinions and not objective (Table 13).

Table 13 Descriptive statistics for subjectivity

```
count    138.00
mean       0.43
std        0.30
min        0.00
25%        0.20
50%        0.50
75%        0.60
max        1.00
Name: Subjectivity, dtype: float64
```

Top 20 positive and negative comments

Using the Textblob method, the top 20 positive and negative comments are provided in Table 14.

Table 14 Top 20 positive and negative comments using Textblob method

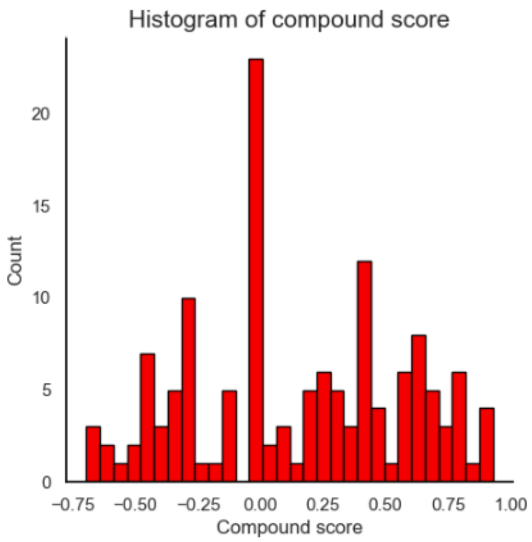
Review	Cluster	Polarity	Subjectivity
TOP 20 POSITIVE			
i would say i am happy	1	0.8	1
servicefirst is a great start for my career learning a lot	1	0.8	0.75
i am happy with sf	1	0.8	1
great place to gain experience	1	0.8	0.75
salary could use an upgrade lol	1	0.8	0.7
servicefirst is giving me great experience	1	0.8	0.75
monthly salary is good	2	0.7	0.6
it is good	1	0.7	0.6
good place to build skills	1	0.7	0.6
it is good cannot complain	0	0.7	0.6
good stepping stone job	1	0.7	0.6
it is a good company	1	0.7	0.6
good pay i guess	2	0.7	0.6
good experience	1	0.7	0.6
no comment it is a good job	1	0.7	0.6
it is good	1	0.7	0.6
getting good experience here	1	0.7	0.6
good income	2	0.7	0.6
servicefirst is a good place to start	1	0.7	0.6
it is good but more performance bonuses	0	0.6	0.55
TOP 20 NEGATIVE			
office politics are the worst	2	-1.0	1.0
the equipment we work with is outdated	2	-0.4	0.6
team collaboration is poor	2	-0.4	0.6
pay is not great	1	-0.4	0.8
been here forever but the pay is still terrible too much travel too	2	-0.4	0.6
company policies are outdated and inflexible	2	-0.4	0.6
i am not happy i have spoken to my team lead about this	1	-0.4	1.0
been here forever but the lack of career growth is frustrating	2	-0.4	0.9
management is not good at all they are not experienced	1	-0.4	0.8
the ceo is not good please change him	1	-0.4	0.6
the workload is ridiculous	2	-0.3	1.0
never any recognition for hard work	2	-0.3	0.5
management can be hard to work with but i like my team	0	-0.3	0.5
hard to balance work and family	2	-0.3	0.5
management is a hard to talk to and unavailable	1	-0.3	0.5
been with servicefirst forever but not sure if i want to continue	2	-0.3	0.9
i hate being away from home so much office all the time is a drag	2	-0.2	0.4
average	2	-0.2	0.4
but the office hours and travel make it hard to have a life outside workj69	2	-0.1	0.3
years of hard work still low pay	2	-0.1	0.4

The top 20 positive and top 20 negative reviews have subjectivity scores above zero reflecting personal feelings rather than objective facts. The common themes for the positive comments are the opportunities for career and skill development and good pay. It was also found that Cluster 1 dominated the positive reviews, which if we recall, had a medium *Satisfaction* score, with mostly young employees early in their career and education (**Low Group**). The common themes for the negative group focused on office politics, poor team collaboration, management inaccessibility, low pay and lack of career growth. In addition, there were complaints about excessive travel, long office hours and difficulty balancing work and personal commitments. The negative comments were dominated by Cluster 2 which were employees that travelled frequently, and were in the medium bracket (medium tenure, income, age and education, **Medium Group**).

NLTK Vader

The histogram of the compound score is provided in Figure 35.

Figure 35 Histograms of compound score



The highest frequency of the comments have been scored as neutral with a much more even distribution either side compared to the polarity distribution using the Textblob method. The mean compound score was 0.16, thus overall positive, with 25% above a score of 0.5 (similar to the polarity score) (Table 15).

Table 15 Descriptive statistics for compound score

```
count    138.00
mean       0.16
std        0.43
min       -0.70
25%       -0.18
50%        0.15
75%        0.47
max        0.93
Name: compound, dtype: float64
```

Top 20 positive and negative comments

Using the NLTK Vader method, the top 20 positive and negative comments are provided in Table 16.

Table 16 Top 20 positive and negative comments using NLTK Vader method

Review	Cluster	Polarity	Subjectivity	Compound
TOP 20 POSITIVE				
my dad used to work at servicefirst and loved it i love it too i am glad it helps families	1	0.6	0.8	0.9
yeah it is pretty good i am happy	0	0.6	0.9	0.9
look sf has been good to me but they seem to reward people with high tenure i am not sure if the junior ones are as happy	0	0.4	0.8	0.9
i am grateful for the high income and the ability to work from home which has improved my overall job satisfaction	0	0.1	0.3	0.9
servicefirst offers excellent benefits and a fixed schedule which allows me to plan my personal life better go sf	0	0.3	0.4	0.8
servicefirst provides a great work environment with minimal travel which has been ideal for my lifestyle	0	0.5	0.8	0.8
great place to gain experience	1	0.8	0.8	0.8
lots of things need to change at sf for me to be happy i am not happy atm	1	0.2	1.0	0.8
i appreciate the high income and the support for remote work which has been crucial for my job satisfaction	0	0.0	0.6	0.8
it is good but more performance bonuses	0	0.6	0.6	0.8
good work sf love working here	1	0.6	0.6	0.8
i love the flexibility of working from home and the consistent schedule which let us me spend more time with my grankids	0	0.4	0.5	0.8
loving the learning opportunities at servicefirst	1	0.6	1.0	0.8
servicefirst is giving me great experience	1	0.8	0.8	0.8
servicefirst has been my home for many years and the opportunity to work remotely has only improved my experience	0	0.1	0.6	0.7
gaining valuable skills	1	0.0	0.0	0.7
sooo the companys commitment to remote work has significantly reduced my travel time making my job much more enjoyable	0	0.3	0.5	0.7
i have stayed with servicefirst for so long because of the excellent pay and the flexibility to work from home	0	0.5	0.7	0.7
having worked here for many years i appreciate the high income and the support for remote work	0	0.2	0.4	0.7
love servicefirst	1	0.5	0.6	0.6
TOP 20 NEGATIVE				
been here forever but the pay is still terrible too much travel too	2	-0.4	0.6	-0.7
stuck with low wages for years management does not seem to care	2	0.0	0.3	-0.7
i hate being away from home so much office all the time is a drag	2	-0.2	0.4	-0.7
office politics are the worst	2	-1.0	1.0	-0.6
low income it is not making me happy	2	0.4	0.7	-0.6
pay is not great	1	-0.4	0.8	-0.6
been here forever but the lack of career growth is frustrating	2	-0.4	0.9	-0.5
the ceo is not good please change him	1	-0.4	0.6	-0.5
it is okay but i think they will have a problem with younger staff and changing environment	0	0.3	0.3	-0.5
team collaboration is poor	2	-0.4	0.6	-0.5
i am not happy i have spoken to my team lead about this	1	-0.4	1.0	-0.5
my team lead needs more experience i am just not happy at the moment	1	0.0	0.8	-0.5
the moneys nice but i am tired of the constant travel	2	0.1	0.7	-0.4
years of hard work still low pay	2	-0.1	0.4	-0.4
low pay more	1	0.3	0.4	-0.4
but too many trips away from home missing out on my kids lives	2	0.2	0.3	-0.4
been with servicefirst forever but not sure if i want to continue	2	-0.3	0.9	-0.4
it is okay cannot think of anything bad that happened to me the last few years	0	-0.1	0.3	-0.4
way too much travel for such low pay	2	0.1	0.3	-0.4
the workload is ridiculous	2	-0.3	1.0	-0.4

It interesting to note that when compared to the Textblob method, the NLTK method seems better in analysing statement sentiment, particularly long statements. For example, the statement, "*I appreciate the high income and the support for remote work which has been crucial for my job satisfaction"* had a polarity score of zero (neutral) but a compound score of 0.8 (positive). Another statement "*my team lead needs more experience i am just not happy at the moment"* also has a polarity score of zero and a compound score of -0.5.

In addition, most of the positive reviews are in Cluster 0 (**High Group**) or 1 (**Low Group),** which included employees that did not travel or travelled infrequently. As with Textblob method, most of the negative reviews were in Cluster 2 (**Medium Group**).

The main themes of the positive reviews included flexibility, remote work, high income and opportunities for skill development. Many employees appreciate a supportive work environment with consistent schedules that improve work-life balance, especially for those with family commitments. Long-tenured employees felt particularly rewarded, while newer staff value the chance to build skills and gain experience.

The main themes for the negative reviews were dissatisfaction with low pay, excessive travel and limited career progression. Leadership issues, such as inexperienced management and poor team collaboration, further exacerbate these concerns. Excessive travel and heavy workloads significantly impact work-life balance, particularly for employees with family responsibilities. These themes highlight a divide between those who benefit from the company's strengths and those struggling with systemic challenges.

Overall, combining both analyses (Textblob & NLTK Vader), positive reviews, mainly from early-career and late career employees in Clusters 0 and 1, emphasise benefits like flexibility, remote work, skill development opportunities and supportive work environments. Negative reviews, predominantly from Cluster 2 employees with medium tenure and frequent travel, focus on challenges such as low pay, excessive travel, poor management and limited career growth. Notably, the NLTK Vader method proved more effective than Textblob in capturing nuanced sentiment in longer statements. These insights emphasise the importance of addressing systemic issues while building on the strengths that contribute to employee satisfaction.

5 Discussion, Conclusion and Recommendation

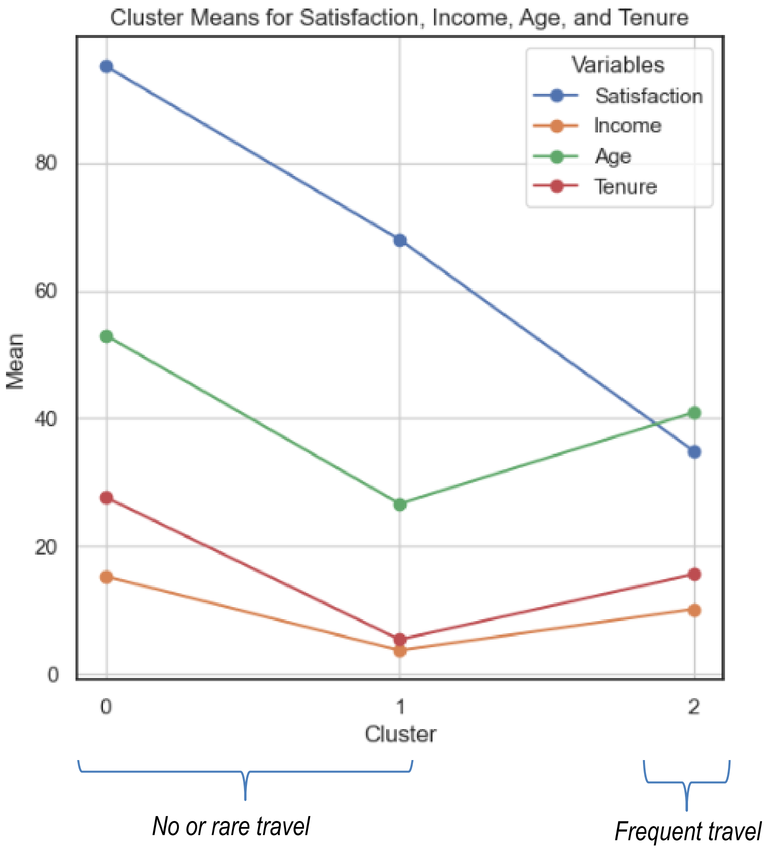
5.1 Discussion

This data science project provided valuable insights into the factors influencing employee satisfaction at ServiceFirst, directly addressing the project's aims. By identifying employee clusters, developing a predictive decision model and conducting sentiment analysis, the findings offer a comprehensive understanding of workforce dynamics and actionable recommendations to improve satisfaction and retention.

The initial data exploration revealed that employees with longer tenures were generally older, had higher levels of education and earned higher incomes. These patterns were consistent across genders, with education showing a strong positive relationship with income, reflecting the benefits of higher education in career progression and compensation.

Satisfaction levels were strongly tied to the extent of travel, with frequent travel significantly contributing to dissatisfaction. Clustering analysis revealed a U-shaped relationship between satisfaction and variables such as monthly income, age, tenure, and education. Three distinct employee groups were identified (Figure 36).

Figure 36 Relationship between Satisfaction, Income, Age, Tenure, Travel and Education



0 = High Group *(high income, age, tenure and education)*
1 = Low Group (*low income, age, tenure and education)*
2 = Medium Group *(medium income, age, tenure and education)*

The **High Group** exhibited the highest satisfaction levels and was characterised by higher income, advanced education, longer tenures and minimal travel demands. These employees may be in senior roles, where their stability and alignment with organisational support systems contribute to their satisfaction. The **Low Group** with moderate satisfaction levels, comprised early-career employees who had shorter tenures, lower education and income levels and limited travel requirements. Their moderate satisfaction suggests that while reduced travel positively influences their experience, limited career opportunities and compensation may temper their overall satisfaction. In contrast, the **Medium group** which demonstrated the lowest satisfaction levels, included mid-level employees with moderate income, tenure and education who travelled frequently. Challenges such as work-life imbalance, travel-related stress and limited career progression may contribute to their dissatisfaction.

The decision modelling supported these findings, with travel frequency emerging as the strongest predictor of satisfaction, followed by the length of employment, age and income. This alignment reinforces the significant impact of travel demands on employee satisfaction, as identified in the clustering analysis. However, the model's effectiveness in distinguishing between employee groups was limited, likely due to its binary categorisation of satisfaction levels. A three-way model aligned with the clustering analysis could improve predictive power by capturing nuanced distinctions among groups.

Sentiment analysis of employee feedback supported these findings. Positive sentiment, from the **High** and **Low Groups**, highlighted benefits such as flexibility, skill development opportunities and supportive environments. Negative sentiment, from the **Medium Group**, centred on issues such as frequent travel, poor management, low pay, and limited career growth. Advanced sentiment analysis techniques, such as NLTK Vader, proved effective in capturing complex feedback, reinforcing the need for targeted interventions to address group-specific challenges.

5.2 Recommendations

ServiceFirst should adopt tailored strategies to address the unique needs of its employee groups to enhance satisfaction and retention. For employees in the **High Group**, who already report high satisfaction, it is essential to implement retention strategies that maintain their positive impact. Recognising and rewarding their contributions, offering leadership development opportunities, and engaging them in mentorship programs may sustain their engagement and leverage their expertise for organisational benefit.

The **Low Group**, comprising entry-level employees, represents an opportunity for growth and development. Enhancing satisfaction for this group may be achieved through structured career progression pathways, competitive remuneration and access to training programs. Supporting their growth not only may build a strong talent pipeline but also foster loyalty and reduce turnover. Additionally, connecting these employees with senior staff in the **High Group** for mentorship may create a sense of belonging and long-term commitment.

The **Medium Group**, which faces the greatest challenges, requires immediate action. Frequent travel is a key source of dissatisfaction and introducing flexible travel policies, such as virtual meeting options and reduced travel requirements, is important. Addressing work-life balance through wellness initiatives, stress management resources and improved career progression opportunities will help mitigate the negative impact of travel demands. Enhancing management practices and providing clear pathways for advancement will further address the root causes of dissatisfaction, improving engagement and loyalty within this group.

5.3 Conclusion

This analysis provides a detailed understanding of the factors driving employee satisfaction at ServiceFirst and highlights opportunities to enhance retention and foster a positive organisational climate. Addressing the distinct needs of each employee group is critical to achieving these goals. Further research is recommended to refine these strategies. Collecting information on job roles, responsibilities and the necessity of travel will provide deeper insights into how role-specific factors influence satisfaction. Investigating the psychological and organisational impacts of frequent travel, such as stress and burnout, may further assist in understanding its effects on performance and well-being. Longitudinal studies are needed to evaluate the impact of implemented strategies over time, tracking changes in satisfaction and retention. Additionally, examining workplace diversity, inclusion, and equity will provide a more holistic understanding of organisational dynamics, while benchmarking satisfaction and compensation levels against industry standards will identify competitive strengths and areas for improvement. By adopting these tailored strategies and incorporating ongoing research, ServiceFirst can enhance satisfaction, improve retention, and foster a more engaged and productive workforce.

6 References

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